

The Impact of Work-from-Home on Brick-and-Mortar Retail Establishments: Evidence from Card Transactions

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Abstract

COVID-19 induced large increases in working-from-home (WFH). We study the impact of these changing work arrangements on brick-and-mortar retail establishments using a novel dataset based on billions of card transactions. We separately identify the effect of residential population movement from the effect of changing workplace locations. While retail establishments follow population movement, more WFH depresses retail establishments. Surprisingly, retail establishments decline more in locations where the residents transition to WFH compared to locations where workers make the transition. Fewer retail trips and substitution to home production are important features for the distinct effects of WFH on retail markets as compared to office markets.

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The modern city relies on the consumption benefits of density (Glaeser, Kolko and Saiz 2001). For the past four decades, high-skilled workers have flocked to downtowns in major cities for both job opportunities and vibrant retail amenities (Diamond 2016; Baum-Snow and Hartley 2020; Couture and Handbury 2020; Couture et al. 2021a). The pandemic has upended that pattern, with work-from-home (WFH) emptying downtown offices and prompting residential moves to suburban neighborhoods and smaller cities (Li and Su 2021; Ramani and Bloom 2021; Althoff et al. 2022; Ding and Hwang 2022; Ferreira and Wong 2022; Brueckner, Kahn and Lin 2023). While the large impact of WFH on office and housing markets is clear, the potentially significant effects for retail markets are less understood and threaten the ability of cities to grow through their consumption benefits (Duranton and Handbury 2023).

The aim of our project is to understand the changing locations and composition of retail establishments due to WFH. For retail establishments, exposure to WFH primarily affects the location and behavior of their customer base, rather than their workforce. Since most shopping trips begin or end at home, WFH may drive retail firms to relocate to where consumers have moved residences (Davis et al. 2019). Additionally, consumers tend to visit retail establishments near their jobs (Miyauchi, Nakajima and Redding 2022; Oh and Seo 2023). Critically, this retail activity near workplaces may not simply move to residential locations if WFH changes demand for retail. For example, a worker could increase their home production by preparing their own lunch rather than visiting a restaurant (Aguilar and Hurst 2007).¹ Thus, the broad impacts of WFH on residential and workplace locations and consumer demand are likely to fundamentally reshape retail markets.

To study the impact of WFH on retail markets, we develop a methodology to track retail establishments using large-scale daily credit and debit card transactions at card terminals. As currently constructed, the dataset we build uses 145 million credit and debit card accounts to track 2.1 million retail establishments in 16 major American cities from Q3 2017 through Q2 2023. Along with the entry and exit dates of each establishment, we track total transactions and spending at each establishment over time. This dataset is uniquely suited to our purpose: high-frequency data capture precise changes in economic activity; disaggregation to the ZIP code level provides rich cross-sectional variation; inclusion of transactions and spending growth measures important economic activity; comprehensive coverage of small, non-employer establishments includes a crucial segment for retail markets; and a unique consumption-based perspective measures the availability of neighborhood retail amenities.

We then document important stylized facts about the changing location and composition of retail establishments from Q4 2019 to Q4 2021. During this two-year period, we first show that retail establishments parallel the exodus of populations from large, expensive cities and city centers for smaller Sun Belt cities and suburbs. At the extremes, the number of San Francisco retail establishments declined by

¹ As documented in [USDA survey results](#) on pre-pandemic WFH food habits.

5.0 percent while Houston's expanded by 5.1 percent. Within each city, downtown retail establishment growth substantially underperformed that in suburban areas. On average, establishments in city centers declined by 2.0 percent while establishments in neighborhoods at the inner and outer suburban rings grew by an average of 2.0 and 0.7 percent, respectively. For context, the Census Business Dynamics Statistics (BDS) report an average two-year growth rate of 2.3 percent for all establishments over the past two decades, with the lowest growth of -3.0 percent in 2010 during the great recession.

The subdued growth in outer suburban retail establishments where population growth is booming suggests that consumers working from home may demand less traditional retail. This is supported by varying impacts across different types of retail establishments. For instance, new restaurant entries are concentrated in inner suburbs while grocery entries increase more in the outer suburbs. This implies that higher concentrations of the WFH population benefit grocery stores at the expense of restaurants as consumers prepare more food at home. Furthermore, leisure establishments are the only other type growing faster with distance from city center. Consumers may continue to demand these establishments despite working from home or demand more of them if they have greater leisure time (Aksoy et al. 2023). This contrasts with the observed decline for clothing, professional consumer services, and personal care services in the outer suburbs, which may be less desired by those who WFH. We also observe persistent suburban decline for establishments in general goods and home goods and services.

Given these stylized facts, we use a simple decomposition analysis to understand the relative effect of changes in residential population growth from changes in workplace locations due to WFH. We proxy for the latter using pre-pandemic exposure to both worker and resident propensity to WFH based on industry composition (Dingel and Neiman 2020; Althoff et al. 2022). The results show that a 1.0 percentage point increase in neighborhood resident population growth is associated with 0.32 percentage points more establishment growth. Then, conditional on resident growth, the adoption of WFH depresses physical retail establishments. A 1.0 percentage point increase in neighborhood pre-pandemic residential or employment exposure to WFH decreases establishment growth by -0.23 and -0.11 percentage points, respectively. Thus, the negative effect of WFH is concentrated around residential locations where many residents were likely to transition to WFH, but also substantial around employment locations where many workers were likely to transition to WFH. The latter effect is driven entirely by employment effects for restaurants. These results support that the relocation and changing product preferences of the WFH population negatively impacts establishments less relevant to their new work environment.

We contribute to research on the impact of WFH on urban real estate markets by showing that there are large effects for retail markets that are distinct from the impact on office markets. Existing literature finds that office-based firms are leaving downtown locations and substantially reducing their commercial office space as their workers transition to WFH (Dalton and Groen 2022; Gupta, Mittal and Van Nieuwer-

burgh 2022). The effects we observe for retail markets are consistent with declining commercial retail rents and valuations in urban areas (Rosenthal, Strange and Urrego 2021; Van Nieuwerburgh 2022). These effects imply spillovers to local labor markets and the ability of local governments to finance public goods through sales and property taxes.

Our results demonstrate that WFH is reshaping the spatial distribution of retail markets through changing the location and retail demand of their customer base. Using national survey data for the U.K., De Fraja et al. (2022) also find large spending declines in areas to which workers commute less during the pandemic. Consistent with our results, Baum-Snow (2020) finds that employment in wholesale and retail trade was among the most likely to relocate in response to highway expansion. These results suggest that incorporating changes in retail markets into spatial models of cities would improve counterfactual and welfare analysis of the rise of WFH (Delventhal, Kwon and Parkhomenko 2022; Delventhal and Parkhomenko 2023; Duranton and Handbury 2023; Monte, Porcher and Rossi-Hansberg 2023; Davis, Ghent and Gregory 2024). Almagro and Domínguez-Iino (2022) and Gorback (2022) are further examples incorporating such endogenous amenities into spatial models.

Our findings also contribute to a long literature on retail location choice, extending back to Hotelling (1929). Subsequent theoretical and empirical work continues to emphasize proximity to customers as a central determinant (e.g., Davis (2006); Ellickson, Grieco and Khvastunov (2020)). Closely related models of entry underscore the role of endogenous product choice at entry (e.g., Seim (2006)). Notable changes in retail goods and service formats in recent decades, such as the shift to online retail and discount stores and the increasing dominance of multi-establishment firms, have also significantly impacted location choice and firm strategy (e.g., Jia (2008); Brynjolfsson, Hu and Rahman (2009); Atkin, Faber and Gonzalez-Navarro (2018); Blevins and Yang (2018); Hsieh and Rossi-Hansberg (2023); Dolfen et al. (2023); Chava et al. (2024)).

Finally, a key contribution of our work is the novel establishment dataset built on transaction data to provide near real-time reliable measurement of retail establishments nationwide. Similar to our approach, Klopach, Lewis and Luco (2024) use card transaction data to assess the impact of Hurricane Harvey on Houston's establishments, demonstrating the utility of such data in another crisis setting. During the pandemic, proprietary data generated in the course of normal economic activity improved the ability of public policy makers to rapidly design and implement programs tailored to the context. For retail markets, transactions and cell phone data tracked changes in spending and visits (Baker et al. 2020; Chetty et al. 2022; Couture et al. 2021b; Relihan et al. 2020; Goolsbee and Syverson 2021). Other work utilized proprietary data to understand the impact on labor markets (Forsythe et al. 2020; Adams-Prassl et al. 2020), small businesses (Fairlie 2020; Crane et al. 2022), and inflation (Jaravel and O'Connell 2020). One hindrance to the public policy response was the inability to directly link consumer spending

declines to the temporary or permanent closure of retail establishments or their subsequent stabilization after policy interventions during a fast-moving crisis. As Klopach and Luco (2024) demonstrate, cell phone-based visit measures are weakly correlated with actual spending, highlighting the greater policy relevance of direct spending measures. Our dataset can fill that gap in the future.²

1 Data

Our primary data source is the credit and debit card transactions for 145 million Chase credit and debit card accounts from January 2017 to December 2023, accessed through the JPMorgan Chase Institute (JPMCI). We focus on the spending in 16 major cities in which Chase has a significant customer footprint.³ In these cities the customer base closely resembles the wider population and includes customers using one or both of debit and credit card products.⁴ As we show in Figure A.1, the unweighted Chase customer base is representative enough to match aggregate population flows from the Census during the pandemic in these cities. In addition, the size of the customer population in these cities increases the probability of observing many transactions at both large and small establishments across every neighborhood on the condition that they accept card payments. The primary data fields we use to construct our measure of establishments are the date and dollar amount of each transaction, whether the card was physically present at the terminal, the location of the terminal (including the ZIP code, city, and country), and the merchant category code (MCC) for the good or service associated with the terminal. We also observe a description of the name under which business is conducted at the terminal.

We limit our study to everyday retail goods and services that are well-captured by card transactions. The following is a non-exhaustive list of the examples of retailers included in each product. Our *General Goods* product type includes department stores, discount stores, internet retailers selling miscellaneous goods, and other retailers that sell non-food everyday goods (e.g. florists or book stores). The *Home Goods* product type includes predominately goods retailers like hardware stores and furniture stores, but also includes some general and specialist contracting services likely employed for home maintenance. Our *Professional Consumer Services* product type includes self-storage, veterinarians, tax preparation, and childcare. *Personal Care Services* includes salons, barbershops, and dry cleaners. *Grocery* includes grocery stores, supermarkets, liquor stores and specialty food retailers. *Restaurants* includes full-service and fast-food restaurants and bars that sell food to be consumed on premises. *Clothing* includes clothing, shoe, and accessory stores. *Leisure* are goods and services including video game stores and venues like movie theaters, bowling alleys, and gyms. *Pharmacy* includes only pharmacies.

²Public datasets are often subject to considerable delays. Points of interest datasets based on cell phone data or open-source tools like SafeGraph or OpenStreetMap may fail to track actual spending or omit smaller, temporary establishments. Visits may not always correspond to transactions, and quick transactions may not be reflected in recorded visits.

³Core Based Statistical Areas (CBSAs) delineate cities.

⁴Ganong and Noel (2019) and Relihan (2024) are examples showing the representativeness of the Chase customer base.

1.1 Measuring establishments with card terminals

Measuring establishments using transaction data for our focal cities and product categories presents several challenges. A key issue is that card terminals do not have a unified establishment identifier linking transactions across multiple terminals and payment processors to the same establishment. To address this, we create a common identifier using the available location and merchant information. However, there are additional complications with the merchant description field. For instance, different card terminals at the same establishment may report variations of the same description, possibly due to being programmed by different people or at different times. Additionally, terminal information may be processed differently depending on the method of payment (e.g., chip vs. swipe) or may be reported differently by different card payment processors. One of the largest sources of variation we observe is that different payment processors allow different character lengths for the text description of the merchant. An example of transactions that could vary in their description but clearly belong to the same merchant, are transactions associated with “Joe’s Famous Pizza” and “Joes Famous Pizz” in the same ZIP code and product category.

Therefore, we develop an algorithm to uniquely identify establishments that resolves the primary sources of description variation:

1. Exclude all online transactions to focus on brick-and-mortar locations where in-person transactions take place.⁵
2. Create clean merchant name and establishment identifiers. Specifically, use regular expressions to distinguish merchant names from establishment numbers in transaction descriptions (if present, such as in “Big Box Store #1234”), remove all remaining non-alphanumeric characters, convert all alphanumeric characters to uppercase, and extend merchant names and numbers to the longest merchant name and number for which it is a shortened version within the same product and ZIP code.
3. Define an establishment as a unique combination of cleaned merchant name, cleaned establishment number, postal code, and product corresponding to a point of sale where in-person transactions take place. Note that establishment numbers enable us to distinguish two establishments owned by the same retailer, such as “Big Box Store 1234” from “Big Box Store 4567” , even if they are in the same ZIP code.
4. Calculate total transaction volume and spending for each establishment at a quarterly frequency.

⁵We classify a transaction as online if that card is not physically present at the point of sale or if the card terminal location information is not a physical location (phone numbers or website addresses recorded as the terminal’s city).

5. Measure the first and last quarter each establishment is observed with at least one transaction and assign those as the entry and exit date, respectively.⁶
6. Exclude establishments that are likely to be noise. This includes establishments observed for only one quarter and those which are observed to be active for less than half of the quarters between their entry and exit dates.

We anticipate ongoing improvements to our algorithm. Currently, it can fail to consolidate some transactions from the same establishment, leading to an overcount. For example, a restaurant with two terminals in which one reports “Joe’s Famous Pizza” and the other “Joe’s Famous Piza” will be counted as two restaurants in a location. While fuzzy matching and machine learning techniques could help reduce this error, inspection of our data and comparisons with public sources suggest that our algorithm successfully reconciles most transactions. We expect these errors to have minimal impact on comparisons across locations or over time, as long as variations in transaction descriptions are random.

Our current algorithm creates a panel of 2.1 million unique establishments based on 39 billion transactions. By share of establishments, grocery, restaurants, and general goods are the three largest products. While we measure establishments in this application on a quarterly basis to enhance the accuracy of tracking very small establishments, implementing higher frequency measurement as needed is straightforward. Our transaction data are available with a short lag and after running our algorithm we exclude the first and last two quarters to improve identification of true entries and exits.⁷ This provides an establishment panel that is less than one-year from the current quarter. Due to the short nature of the panel, we do not seasonally adjust our statistics. Instead, we report any growth over time as one- or two-year-over-year changes using the same calendar quarter from the base year.

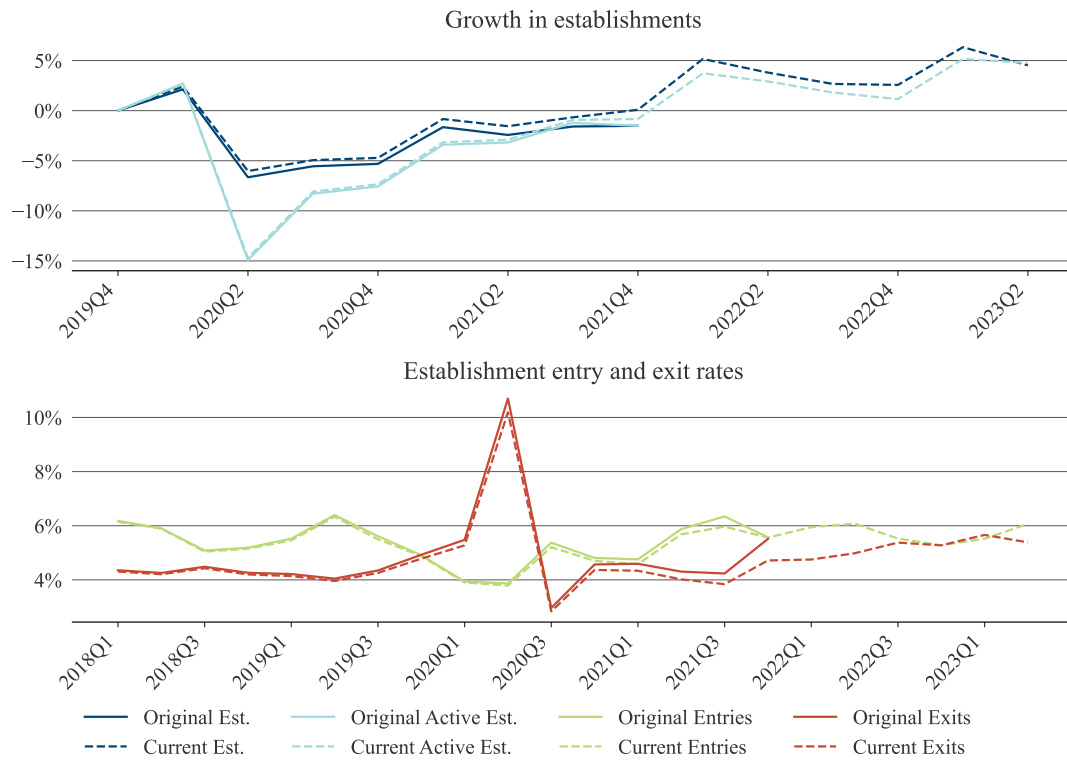
With this resulting dataset, we trace out aggregate growth and the quarterly entry and exit rate of retail establishments in Figure 1. The top panel shows aggregate establishment growth relative to 2019, differentiating between active establishments (at which there are transactions in that quarter) and all establishments. The bottom panel shows quarterly entry and exit rates. Both panels show measurements with data ending in our original estimate as of Q4 2021 and our latest estimate as of Q2 2023. While the current estimate shows modest downward revision of exit rates in 2021, this provides confidence that removing the first and last two quarters from our analysis is sufficient for measuring true entry and exit.

The figure reveals key patterns in establishment dynamics over time. The surge in the exit rate due

⁶Thus, seasonal establishments like ice cream parlors are inactive in winter rather than non-existent for quarters between their entry and exit dates.

⁷By construction, all establishments that exist prior to Q1 2017 or after Q4 2023 have these end dates defined as their entry or exit dates, respectively. We also exclude an additional quarter at the beginning and end of the panel because some establishments have too little activity in the first or last quarter to be detected by terminals, either through the size of the establishment or the seasonal nature of its product. This would artificially inflate the entry and exit rates for Q2 2017 and Q1 2022, respectively. Similar effects in subsequent quarters may show similar effects, but to a decreasing extent away from panel end dates.

Figure 1: Aggregate changes in establishments across data vintages



Notes: The top panel shows growth in all establishments versus active establishments relative to the same quarter in 2019. Active establishments are those at which customers make transactions in the quarter. The bottom panel shows the entry and exit rate relative to the stock of establishments in the previous quarter. Measurements made using data through June 2022 from our original measurement are compared to those made using data currently through December 2023.

Sources: JPMorgan Chase.

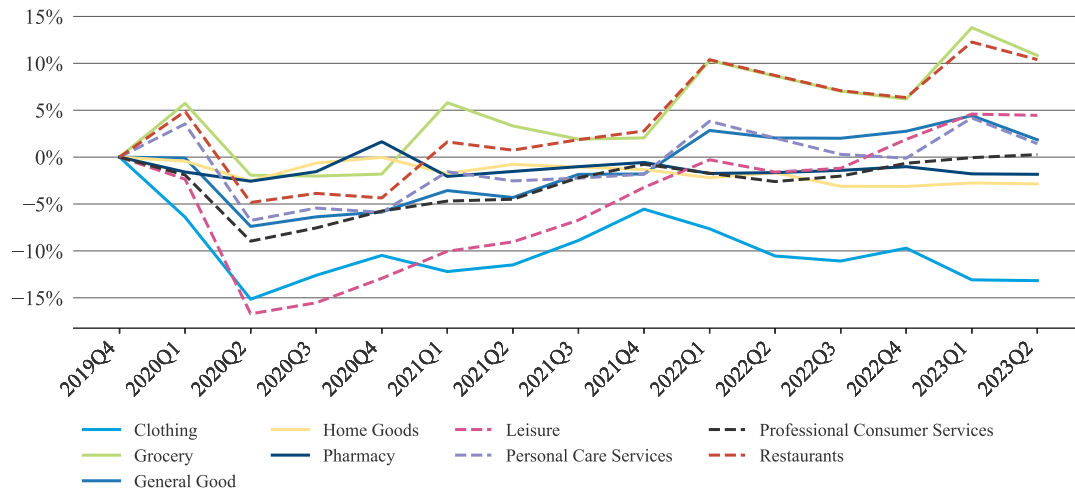
to the pandemic was limited to Q2 2020 and dropped below the 2019 average in Q3 2020. This suggests that the pandemic partly accelerated closures that were already likely to have taken place. By Q4 2020, exit rates had returned to their 2019 level, while the entry rate of new establishments in the second half of 2020 remained muted. By the end of 2021, both entry and exit were higher than their pre-pandemic averages. As we show in Figure A.12, recent entry and exit is highly spatially concentrated, suggesting that spillover effects between retailers may partly explain continued high entry and exit rates. By Q4 2021, the number of total establishments had fully converged, with no meaningful difference between active and all establishments. As of Q2 2023, there are 4.5 percent more establishments than in 2019.

While we provide more detailed comparisons with existing data sources below, initial comparisons of these aggregate measures are informative. The quarterly rates of establishment entry and exit we observe in our data in 2018 are an average of 5.6 and 4.2, respectively. This is above the 2.2 and 2.1 quarterly entry and exit rates for metropolitan and micropolitan areas implied by the yearly BDS. The higher rate may be related to the differences in geographic and industry coverage as well as more representation of small establishments who enter and exit at higher rates. Consistent with Crane et al. (2022) and Fairlie (2020), we find that establishment exit was much more muted than was feared given

the declines in consumption in the early pandemic. Our measure shows that even though 15.2% of establishments were completely inactive in Q2 2020, only one-third of that decline was due to permanent closures.

Figure 2 shows aggregate establishment growth patterns by product. Groceries and restaurants exhibit the strongest growth relative to the pre-pandemic baseline by the end of our sample. While clothing, general goods, personal care, and leisure were the weakest performers in Q4 2021, clothing and general goods continue to decline, whereas personal care and leisure show signs of recovery.⁸ The recent poor performance of clothing aligns with substantial declines in spending since Q4 2021. Appendix Figure A.9 illustrates that, by July 2022, 13.8 percentage points fewer customers in a stable panel shopped for clothing offline each month, and online clothing spending was 26.0 percent lower relative to the pre-pandemic baseline.

Figure 2: Aggregate changes in establishments by product



Notes: This figure shows growth in establishments by product with growth relative to the same quarter in 2019.

Sources: JPMorgan Chase.

One concern with our measure during the pandemic is that more establishments could have adopted card payment terminals for the first time early in the pandemic than in a typical quarter. This could create a surge in false entries. While we are not aware of data that tracks payments accepted by merchants in the U.S., the [Merchant Acceptance Survey](#) of small and medium retail businesses (50 or fewer employees) from the Bank of Canada shows that since the pandemic, acceptance of credit and debit cards among Canadian merchants increased from approximately 67 to 88 percent. Virtually all larger merchants already accepted card payments. Given that a lower share of the population is unbanked in urban areas in the U.S., pre-pandemic use of card terminals by merchants in our context was likely higher ([Benson, Grundl and Windle 2020](#)). In Figure 1, we observe that the entry rate of establishments in Q3 2020

⁸Crane et al. (2022) also provide establishment closure statistics by subsector, showing similar results. Like our findings, they report that permanent restaurant closures were far smaller than initial spending declines suggested, and that leisure-related establishments were among the hardest hit during the pandemic.

increased by 1.7 percentage points before moderating again in Q4 2020. Some of this increase may be false entry due to terminal adoption. However, we do not observe a higher entry rate for small independent establishments compared to chain establishments, where we would expect a gap if smaller merchants were more likely to adopt card payment terminals (Figure A.3). This suggests that the impact of card payment adoption on our measurement during the early pandemic is minimal.

1.2 Comparison with public data on retail good and service establishments

To validate our establishment measure, we compare it by product with three publicly available data sources. The first is the BDS, which is based on the Census Bureau’s Business Register of employer companies and is considered a definitive source for establishment dynamics ([Haltiwanger, Jarmin and Miranda 2013](#)). The second source is the Quarterly Census of Employment and Wages (QCEW) compiled from employer participation in state unemployment insurance programs. Similarly comprehensive in coverage, the QCEW provides data at a quarterly frequency and for specific county and industry combinations. The third is the County Business Patterns (CBP), compiled from the Business Registrar and complementary data from other administrative records and surveys in March of each year. Though explicitly not designed for tracking establishment dynamics over time, the CBP is the only source of annual disaggregated counts of establishments by industry at the ZIP code level. In our comparisons, we focus on three product categories that align closely, though still imperfectly, with the North American Industry Classification System (NAICS): grocery, restaurants, and personal care services.⁹

Overall, comparison to these sources suggests sample coverage and construction differences that advantage our measure for retail good and service establishments. For instance, we find the closest alignment with these three public sources in the restaurant product category, which makes sense as nearly all restaurants accept card payments and also have employees and report payroll, making it likely that most appear in both sources. This implies that in other product categories where we show more differences, some of that difference is likely due to very small non-employer retail establishments that may not report payroll but still accept card payments.¹⁰ These could include small corner bodegas, family-run restaurants, and sole-proprietor hair salons. Capturing them is important for retail markets as they are plausibly more common than other industries, like manufacturing. They are also likely to enter and exit at higher rates, leading to persistent level differences between our measure and public sources.

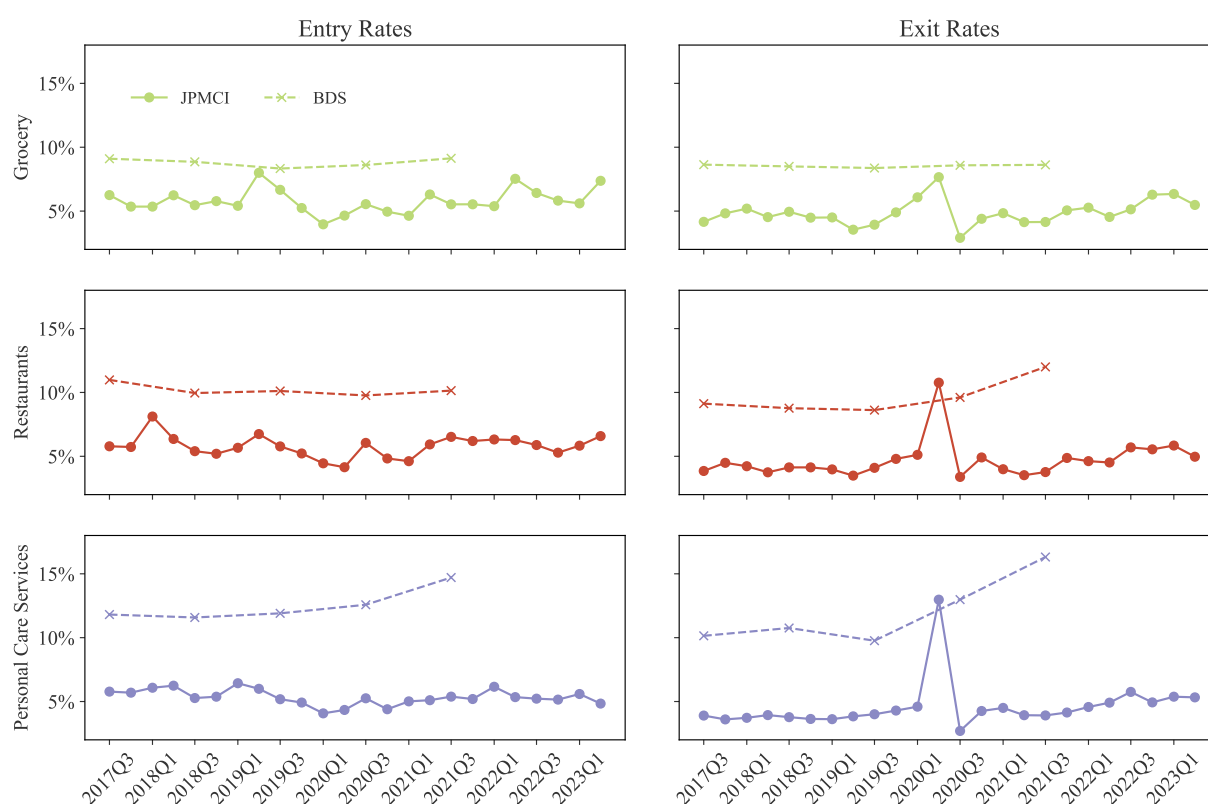
Differences in coverage can also arise because our measure is a consumption-based, rather than production-based, view of where retail establishments are located. This manifests, for example, in the farmers market stands we observe that generate many card payments but may not be recorded as

⁹ Appendix Section 5.2 discusses our mapping to NAICS codes.

¹⁰ [Barnatchez, Crane and Decker \(2017\)](#) review research documenting a similar phenomenon when comparing Census to the National Establishment Time Series based on the privately collected Dunn & Bradstreet data.

employment locations. Other establishments, like bakeries and general merchandisers, can generate sales for grocery and other products that give a fuller view of where groceries are available, but also generates more establishments in our measure than in datasets that singly classify establishments. The employment structure at an establishment can also play a role. For instance, in personal care services, what may be elsewhere categorized as a single salon can appear to us as a collection of individual personal care service providers if the stylists rent and operate their chairs independently. We view these differences as salient for interpretation, but not necessarily mismeasurement, and valuable in trying to understand consumption, rather than production, locations. We now make a comparison with each data source in more detail.

Figure 3: Comparison to BDS entry and exit rates



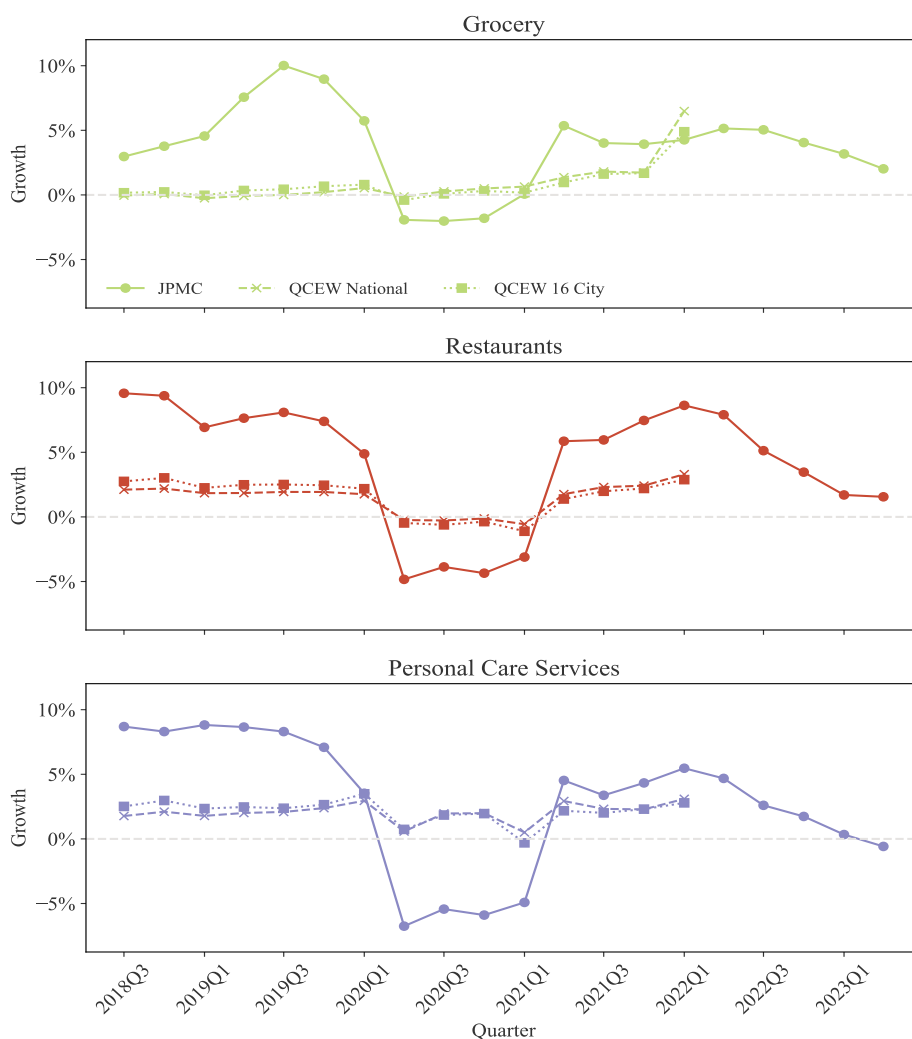
Notes: This figure shows the correspondence between yearly entry and exit rates in the Census BDS and quarterly entry and exit rates in the JPMCI measure.

Sources: JPMorgan Chase, U.S. Census Bureau Business Dynamics Statistics.

BDS: Figure 3 compares our quarterly entry and exit rates with yearly national entry and exit rates reported by the BDS. While the quarterly JPMCI rates are lower than the annual BDS rates, they are higher than the BDS's implied quarterly rates. Despite these level differences, both series show similar patterns in timing and scale of changes across products. The left column indicates steady entry rates through the pandemic according to both sources. An exception is observed for personal care services, for which the BDS shows more pronounced increased entry rates in 2020 and 2021. In the right column, the JPMCI measure shows sharp increases in exit rates in Q2 2020 which match the relative severity

of exits by product reported by the BDS for 2020 and 2021. However, the BDS data reveals a gradual increase in exit rates over these years, unlike our more immediate spikes. This difference stems from our measurement focusing on the cessation of customer transactions, while the BDS would capture a slower wind-down of employment and other activities. A reversal of this lead-lag relationship is plausible for entry. We also show that in the most recent data for 2022 and 2023, entry and exit rates exhibit steady increases.

Figure 4: Comparison to QCEW growth rates



Notes: This figure shows the correspondence in establishment growth rates between the BLS QCEW and the JPMCI measures. In addition to the national QCEW rate, the figure shows the aggregate for the 16 cities included in the JPMCI measure. QCEW data for each city is the aggregated county data corresponding to each city. The largest counties accounting for a cumulative 90% of city population are included.

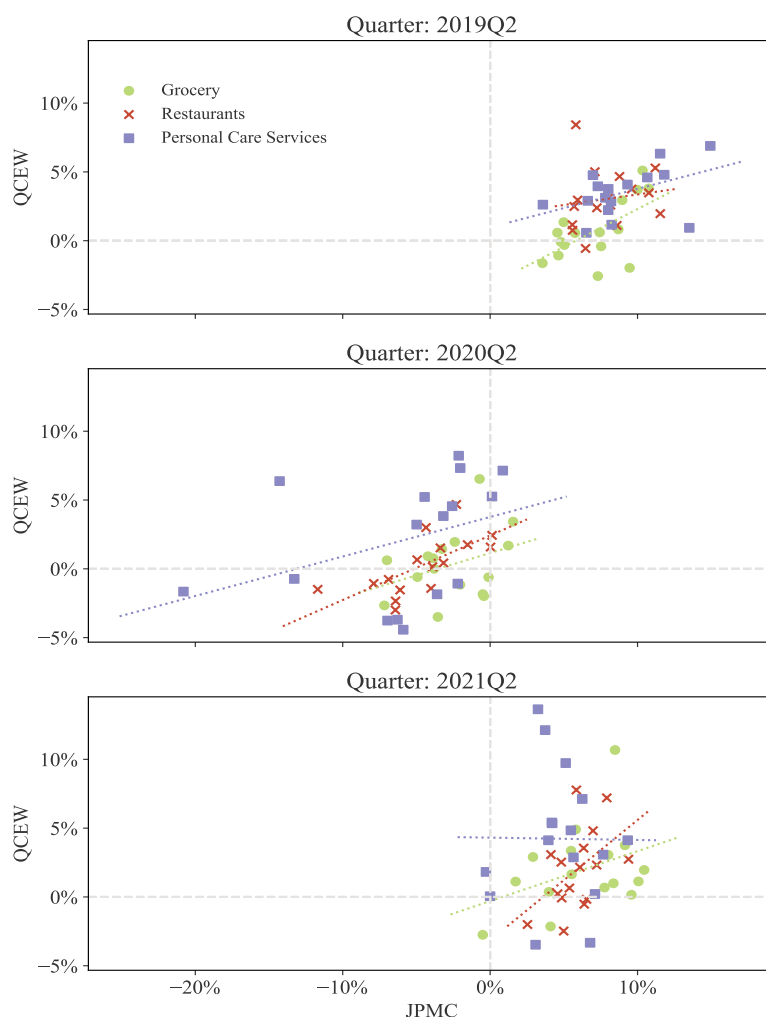
Sources: JPMorgan Chase, U.S. Bureau of Labor Statistics Quarterly Census of Employment and Wages.

QCEW: We compare our establishment growth to the QCEW in the aggregate and by city and show close tracking of quarterly growth rates over time and across locations. Figure 4 shows a time series comparison for each product, comparing the JPMCI measure against both the national and same 16-city QCEW growth rates. The timing and magnitude of downturns are similarly captured by both measures,

with the JPMCI measure identifying the onset of pandemic-related downturns as early as Q1 2020. While the QCEW data reveal little variation between national and the 16-city growth rates, the QCEW shows that these 16 cities experienced faster establishment growth than the national average before the pandemic. The relative growth reversed post pandemic, with these cities growing slower than the country overall. Slower growth post-pandemic is the same pattern observed in our measure.

Figure 5 demonstrates a consistent pattern in identifying which cities are experiencing more significant growth or decline, both before and during the pandemic. Each subplot is a scatter plot of city growth rates by industry in our measure and the QCEW. The only substantial disagreement is in the Q2 2021 growth rates for personal care services, where the QCEW shows some outlier cities with much higher growth rates. This is likely driven by entry, as suggested by the discrepancy in entry rates already observed in Figure 3.

Figure 5: Comparison to QCEW growth rates across cities

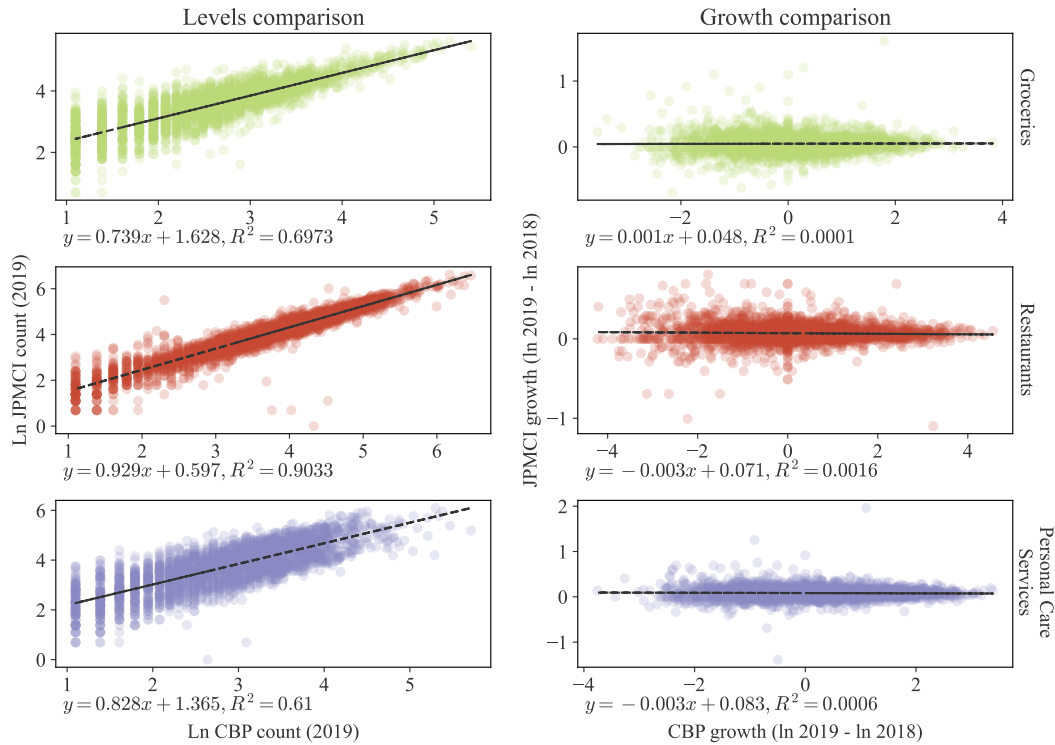


Notes: This figure shows the correspondence in establishment growth rates for select quarters by city between the BLS QCEW and the JPMCI measure. QCEW data for each city is the aggregated county data corresponding to each city. The largest counties accounting for a cumulative 90% of city population are included.

Sources: JPMorgan Chase, U.S. Bureau of Labor Statistics Quarterly Census of Employment and Wages.

CBP: In Figure 6, we compare the count and growth of establishments by ZIP code, using JPMCI data for Q1 2018 and Q1 2019, and CBP data for 2018 and 2019. As shown in the left column for 2019, the JPMCI measure closely aligns with the CBP on establishment locations, particularly for restaurants.¹¹ However, our measure consistently identifies more establishments than the CBP across all three product categories, as indicated by the regression lines with intercepts above zero. While some overcounting may stem from our algorithm not fully consolidating transactions to each establishment, this alone would likely lead to similar levels of overcounting across products. The larger intercepts for grocery and personal care services compared to restaurants suggest that overcounting is more prevalent in products with more small and non-employer establishments. Therefore this overcounting may, instead, reflect greater coverage of these segments.

Figure 6: Comparing to the CBP in levels and growth



Notes: In the left column, this figure shows establishment counts by ZIP code for selected products in the JPMCI sample in Q1 2019 as compared to the CBP in 2019. The right column shows the difference in log growth rates from Q1 2018 to Q1 2019 for JPMCI and 2018 to 2019 for CBP.

Sources: JPMorgan Chase, U.S. Census Bureau County Business Patterns.

In the right column of Figure 6, the comparison of growth rates shows no agreement on which ZIP codes are increasing or decreasing in establishments over a one-year horizon. This discrepancy likely partly reflects differences in coverage, as seen with the BDS and QCEW comparisons. However, another important contributing factor in this case is that the CBP provides successive cross-sectional snapshots without tracking of the same establishments over time. Given our strong alignment in dynamics with

¹¹We exclude the nearly identical 2018 comparison for brevity.

BDS and QCEW, which do track establishments over time, the CBP’s growth discrepancies suggest that our transactions-based measure offers a significant advantage in tracking establishment dynamics at finer spatial and temporal scales.

In the appendix Section 5.4, we further compare our measure of establishments to the CBP. We show that for restaurants, excluding the smallest establishments brings our measure into closer alignment with the CBP. Additionally, we find that the differences between JPMCI and CBP are more pronounced in areas with unique characteristics, such as ski resorts, universities, and farmers markets, where our measure captures more temporary or small-scale establishments. These observations further support the robustness of our measure in detecting smaller and less typical retail locations, which are often excluded by employment-based data sources like the CBP.

2 Stylized facts: Q4 2019 to Q4 2021

We now utilize our establishment measure to study the effect of the pandemic-driven shift to WFH on the changing locations and composition of retail establishments. This analysis focuses on a two-year window from Q4 2019 to Q4 2021, as our data on customer residential locations as of this writing is available only through 2021. While this limits our ability to jointly analyze population and establishment changes beyond that point, this period is still well-suited for assessing WFH’s impact on retail markets. By the end of 2021, in-person visits had recovered from pandemic lows, so store entries and exits likely reflect lasting shifts in customer bases (Figure A.9). Additionally, restricting the study to this period minimizes the influence of broader equilibrium effects, such as residential relocations driven by WFH-induced retail changes. We begin by presenting a set of stylized facts on population and establishment trends across space during this time.

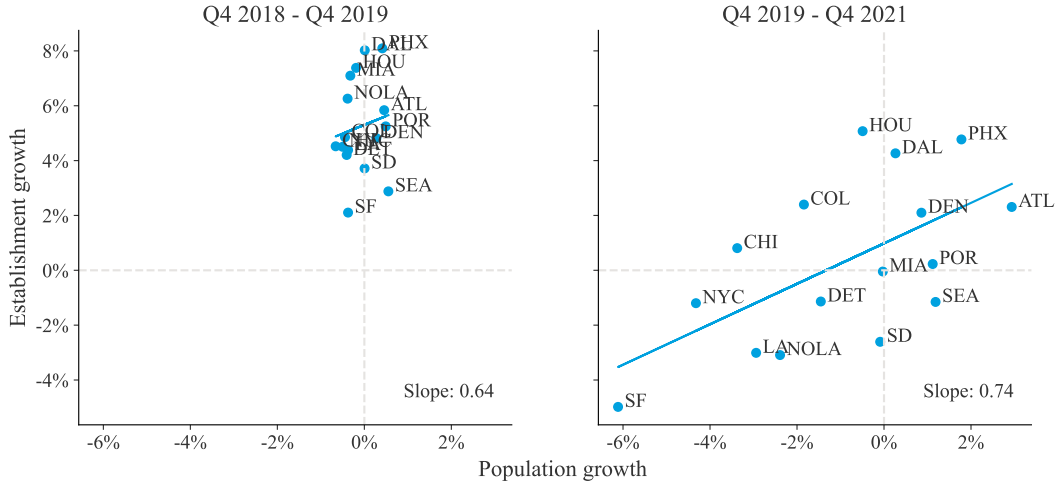
2.1 Establishment and population growth across cities

We first compare aggregate city establishment growth against population growth, where population growth is measured with changes in reported addresses of Chase customers.¹² The results in the left panel of Figure 7 show that in the year before the pandemic, establishments increased in all cities despite already declining populations in many high-cost coastal cities. In the first two year after the pandemic, there is wide variation across cities in establishment recovery that more strongly reflects population changes during the pandemic. Cities such as San Francisco, Los Angeles, San Diego, and New York City experienced some of the largest declines in establishments in addition to large population declines.

¹²Customer address ZIP codes are tracked on a monthly basis. When a customer reports multiple ZIP codes within a quarter, we use the modal ZIP for the quarter. To be included in the cohort for the population growth calculation between two quarters, customers must have 10 transactions in each month of both quarters. This limits error from inactive customers who are less likely to report changes in address and the effects from entry and exit from the Chase customer base.

The stronger correlation with population declines during the pandemic may be the result of the changing demographics of movers toward the high end of the socioeconomic spectrum (Li and Su 2021). These movers would comprise the core of many retail establishments' customer base.

Figure 7: Establishment and population growth across cities



Notes: The left panel shows the one-year establishment and population growth correlation prior to the pandemic while the right panel shows the two-year correlation during the pandemic. We show the one-year correlation prior to the pandemic because we start our quarterly panel in Q1 2017 and cannot measure year-over-year quarterly growth until Q3 2018. See our discussion of the panel construction in our appendix for further details.

Sources: JPMorgan Chase.

2.2 Population and establishment growth within cities

To facilitate within-city comparisons, we normalize distance between each ZIP and the center of its corresponding city along a 0 to 1 distance scale. We follow Holian (2019) and Couture and Handbury (2020) in defining the city center based on the location of the dominant city's city hall.¹³ Each scaled distance is then the ZIP centroid distance from city center divided by the distance of the farthest ZIP centroid in the city. For exposition, we define neighborhoods in the downtown core as those ZIPs from a distance of 0.0 to 0.1, the outer core neighborhoods as those from a distance of 0.1 to 0.4, the inner suburbs as neighborhoods from a distance of 0.4 to 0.8, and the outer suburbs as those from 0.8 to 1.0.¹⁴

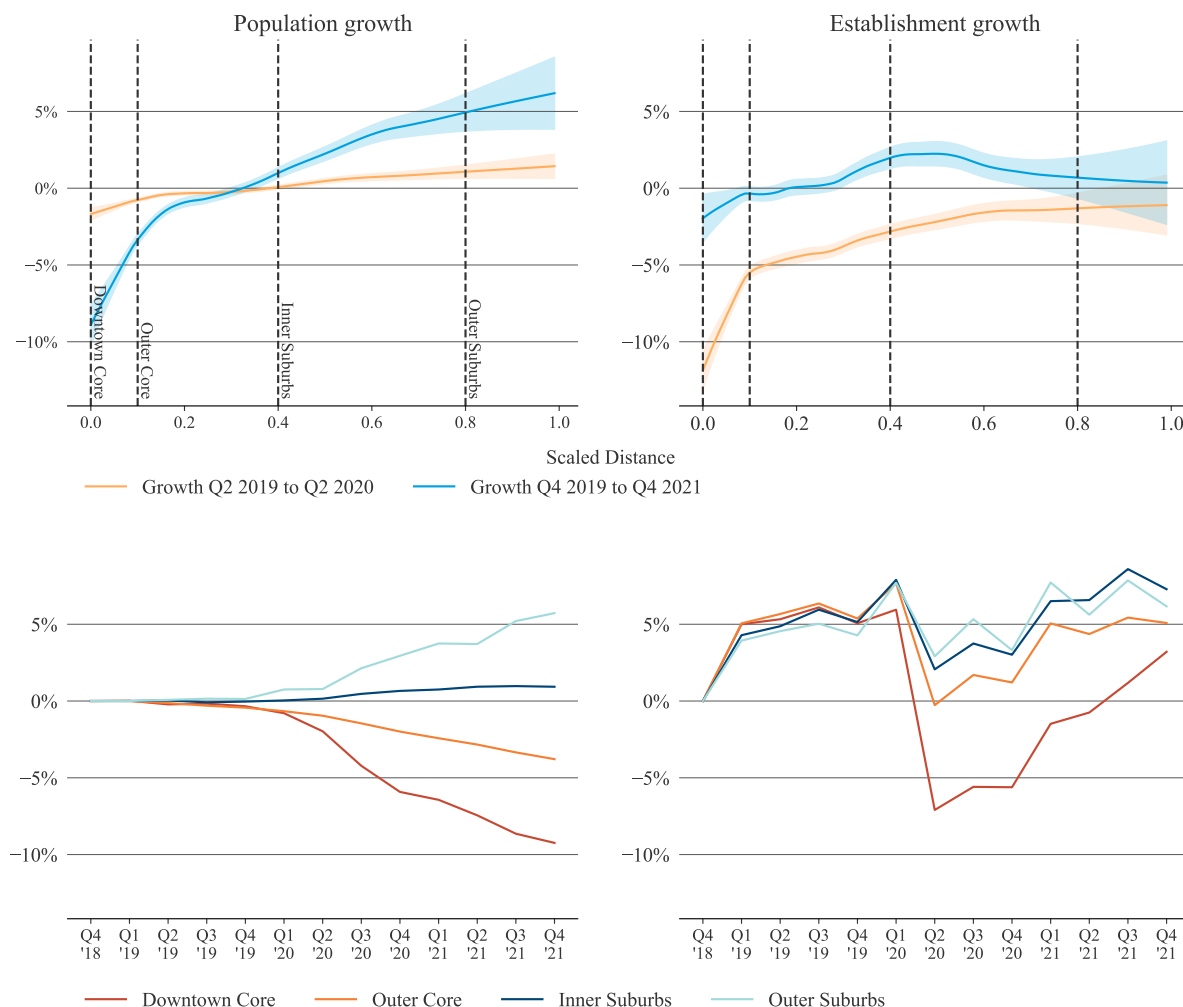
We then estimate average establishment and population growth within cities at each distance using locally weighted scatterplot smoothing (Cleveland 1979). For these regressions, our dependent variables, ΔEst_{zq} and ΔPop_{zq} , are the growth rates in establishments or population, respectively, in ZIP z between quarter q and the same quarter in 2019. Our independent variable, $dist_z$, is the scaled distance from city center. Smoothed values of the dependent variables in each quarter are estimated by taking a subset of the points closest to $dist_z$ in distance (those within the span), and using a kernel function to specify

¹³We take the pre-1954 city hall location in the central business district for Miami.

¹⁴Figure A.11 shows city maps with these neighborhood regions.

weights for those points in a weighted linear regression. A span of one-third of the data and the standard tricube weighting function are used in our estimation.

Figure 8: Population and establishment growth within cities



Notes: Scaled distance is calculated as the distance between the centroid of each ZIP and the city hall of the principal city of its CBSA, then dividing this distance by the greatest ZIP to city hall distance in the CBSA. In panel (a), we index growth for both population and establishment growth for the early (Q2 2020) and late (Q4 2021) to the corresponding quarter in 2019 to abstract from seasonality differences in growth. Bootstrapped 95 % confidence interval bands are shown as shaded areas in the top panel. In panel (b), growth in each quarter is relative to the same quarter in 2018.

Sources: JPMorgan Chase.

We find that early pandemic conditions created different growth curves for populations and establishments within cities.¹⁵ This can be seen in the orange lines in the top panel of Figure 8. At the start of the pandemic in Q2 2020, population declines in the downtown and outer core areas were modest and similar to previous quarters. We observe little immediate effect for population because initially temporary customer moves out of the city would not yet be reflected in permanent address changes. However, our measure of establishments shows immediate permanent closures of retail establishments across the city, with greater declines toward the city center. For neighborhoods at the center, there is an average

¹⁵Decker and Haltiwanger (2022) also document stronger business formation in suburban areas since the pandemic.

effect of 11.9 percent fewer establishments relative to Q2 2019. Moving marginally out of the city center to neighborhoods at a distance of just 0.1 dramatically reduces the negative effect to just 5.5 percent. From there, the gradient toward the outer areas of the city is much less steep. Neighborhoods farthest away from the center experienced just a 1.1 percent drop in establishments at the start of the pandemic.

There are important differences in these growth curves as of Q4 2021. The first noticeable feature is the dramatic rotation of the population curve. Neighborhoods at the city center and the start of the outer core declined 8.9 percent and 3.4 percent, respectively. Population growth at the starting points for the inner and outer suburbs, however, was robust at 1.0 percent and 4.9 percent, respectively. The time series in the bottom panel of Figure 8 shows that this pattern is the result of a continuous population shift from downtown to suburban areas over the previous two years. This shift is decelerating but continued to grow through the end of 2021.

In contrast, our results for establishments show a concave growth pattern. For Q4 2021, establishments in city centers declined by 2.0 percent, increased up to 2.2 percent in the inner suburbs, and then experienced weaker growth in the outer suburbs. The bottom panel illustrates that downtown cores were still recovering from their initial decline, while suburban areas maintained steady growth rates from Q1 2021. Additionally, the bottom panel indicates that the year prior to the pandemic did not exhibit this concave growth pattern, and all areas experienced comparable establishment growth. The recent weak establishment growth in the outer suburbs, despite substantial population growth, suggests that the concentration of people working from home may dampen retail activity.

2.3 Establishment growth for different goods and services

To explore these city-wide patterns, we replicate our within-city establishment analysis by product type. For this analysis, we use half the data in the span and the same tricube weighting function. Figure 9 shows the results for establishment growth, complemented with transactions growth at these establishments to better understand changes in realized shopping trips that could drive establishment growth.¹⁶ The most striking difference is that while transaction growth consistently increases with distance for every product, this has not uniformly translated into higher establishment growth. This indicates that in product categories where establishments have declined but transactions have risen, more transactions are concentrated at a smaller number of establishments.

To date, there are only two clear categories of retail establishments that grow with distance. The first is grocery. Moving from downtown to the suburbs, growth was continuously stronger and more robust than for any other product anywhere within cities. The second category is leisure establishments (e.g. gyms and theatres). Demand for these establishments is likely more independent of where and how

¹⁶Figure A.4 shows qualitatively similar patterns for spending.

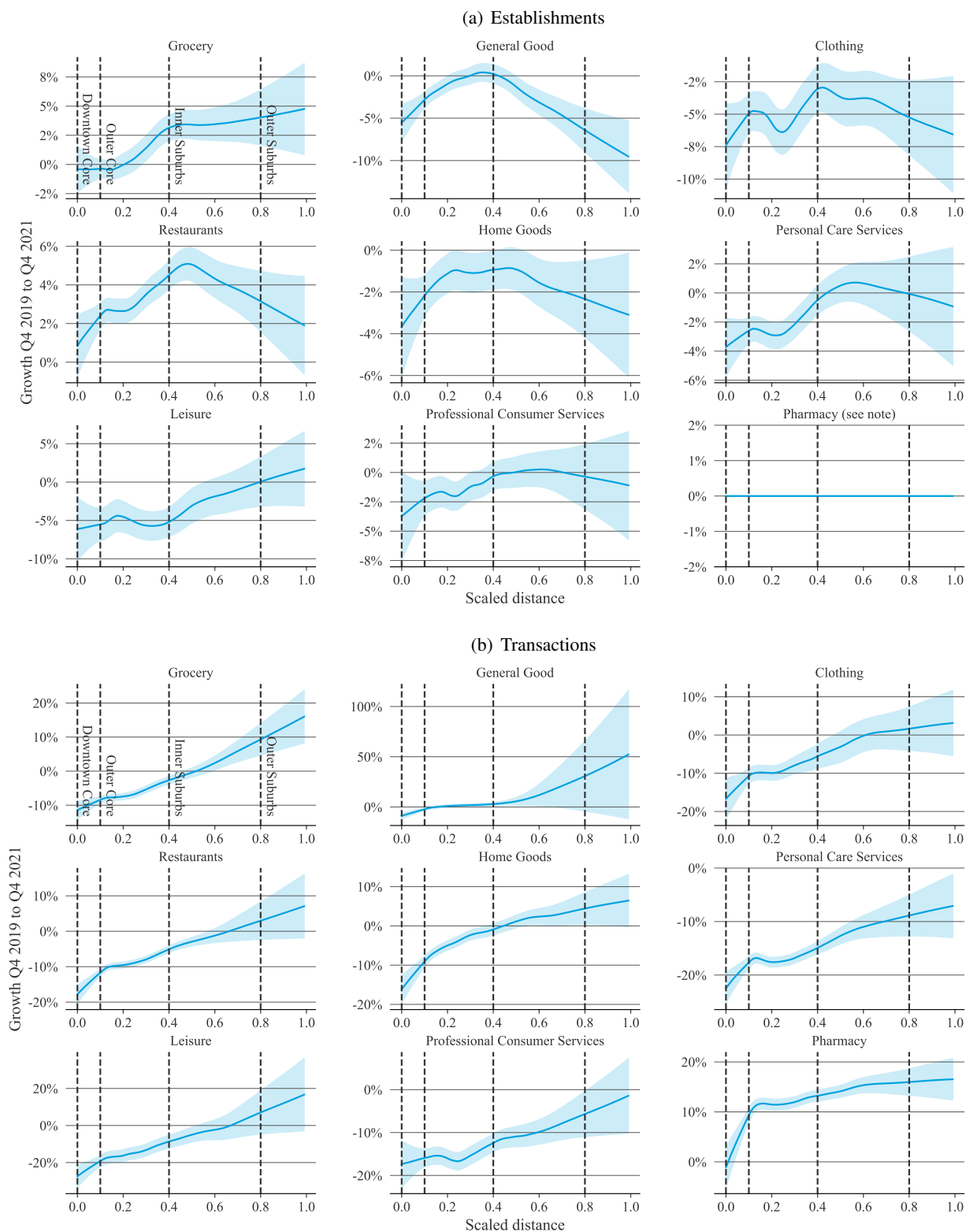
residents work and can increase as WFH residents spend more time at home (Aksoy et al. 2023).

Other products show positive growth in establishments concentrated in the inner suburban areas of cities. For example, restaurants' strongest growth is 5.0 percent at the midpoint. This may reflect stronger shifts in demand from restaurants to grocery stores farther into the suburbs, where more of the WFH population has moved. The widening gap in transaction growth between grocery and restaurants from the inner to outer suburbs further supports this substitution: at a scaled distance of 0.4, grocery transactions are only 2.4 percentage points higher, but this gap steadily increases to 8.9 percentage points at the city edge. Zoning restrictions or higher minimum customer density requirements may also prevent restaurants from equally following residents into the outer suburbs. Patterns for clothing, personal care services, and professional consumer services show similarly weaker growth from the inner to outer suburbs, consistent with reduced demand from individuals who moved to the suburbs during the pandemic or already lived there and shifted to WFH.

The declines almost everywhere within cities for general goods, home goods, and clothing are somewhat surprising. For instance, average transactions growth in the suburbs was higher for general goods than for any other product type and people who WFH might be expected to demand more home goods products. What might explain this disparity in the case of general goods is that these establishments are dominated by big-box and department stores, establishments with large physical footprints which are often custom-built by large chain retailers. They may therefore be slower to respond to this demand shock in terms of establishment locations. But for these goods products, a further contributing factor could be online retail competition. The pandemic may have accelerated the closure of many establishments unlikely to be profitable given online retail's continued growth. Indeed, in the appendix Section 5.5, we show that online spending growth for goods products during the pandemic was strong and that home goods and clothing are the only two products that show declines in overall spending offline after the rebound from the acute phase of the pandemic.

Finally, pharmacies show no differential growth rate within cities so far. This could reflect broad transactions growth at these establishments everywhere within cities, limiting exit, and licensing requirement for new locations, limiting entry. The dominance of large chain pharmacies with long-term leases could also contribute.

Figure 9: Establishment growth within cities by product



3 Empirical estimation

The patterns discussed in Section 2 suggest that the impact of WFH on retail goods and services establishments within cities operates through three main channels. First, the relocation of residents from downtown to suburban areas shifts residence-based consumption toward the suburbs. Second, the shift to working from home moves consumption related to employment from central business districts to residential areas, as consumers now work from their homes. Third, consumers who adopt WFH may prefer a different bundle of goods and services, leading to changes not only in where they live and work but also in their overall demand for retail products.

Identifying the independent effect of each of these channels is challenging based on these stylized facts alone. In addition, variation across neighborhoods and product types — such as capital costs, zoning restrictions, lease structures, customer density, and competition from online retail — suggest further complexities. These factors necessitate a unified empirical analysis to isolate the channels through which WFH affects retail markets, while controlling for these and other potential confounders.

3.1 Estimating equation

To measure the impact of WFH, we would ideally measure which customers are engaged in remote work and then directly study their residential moves and spending changes near their residential and employment locations. However, there is no direct WFH or employment location indicator for individuals in our data. Outside data for this purpose is also limited. Therefore, we utilize second-best measures which serve as proxies for each of these channels.

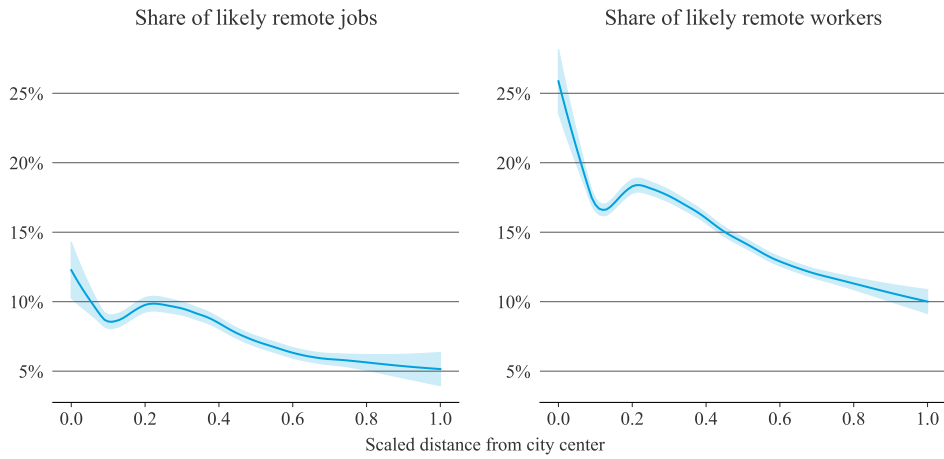
To capture the effect of residential moves induced by work from home, we use population growth from Q4 2019 to Q4 2021, which we measure directly from the residential moves of Chase customers. This assumes that population growth over the two-year window is driven by the moves of residents who have transitioned to work-from-home. While some of the residential moves during this period are, of course, for other reasons, the sharp break in population growth rates prior to the pandemic and the pattern of movement from downtown to suburban areas strongly suggests that work from home drove the majority of these residential moves in this period.

To then capture the impact of WFH adoption, conditional on residential moves, we utilize standard measures of pre-pandemic exposure to WFH in each ZIP code. To build our measure of WFH exposure, we collect data on the industry composition of both residents and workers from the 2019 Longitudinal Employer-Household Dynamics Local Origin-Destination Employment Statistics (LODES). Employment exposure is measured by the percent of employment located in a neighborhood in industries that had a high likelihood of transition to fully remote or hybrid working schedules. Residential exposure

is measured by the percent of residents in industries that had a high likelihood of transition. These industries are the two-digit NAICS industries corresponding to Information (51), Finance and Insurance (52), Professional Services (54), and Management of Companies (55) and are closely correlated with the adoption of remote work during the pandemic (Althoff et al. 2022; Dingel and Neiman 2020).¹⁷

Figure 10 shows a smoothed scatterplot of the WFH employment and residential exposure across scaled distance for the 16 cities we study. Along both margins, downtown cores were more than twice as exposed as areas at the city edge. Furthermore, in this sample of large cities, it was generally the case that more of the workers than the jobs in each location were exposed to WFH. The non-monotonicity from 0.1 - 0.25 scaled distance is created by the water surrounding the downtowns of New York and San Francisco at those distances.

Figure 10: WFH employment and residential exposure by distance



Notes: This figure shows a smoothed scatterplot of the percent of jobs and workers likely to transition to remote work in ZIP codes by scaled distance from city center. Remote work-exposed industries are in NAICS codes 51, 52, 54, and 55 as in Althoff et al. (2022). Bootstrapped 95% confidence intervals are shown in the shaded region.

Sources: JPMorgan Chase, U.S. Census Bureau Longitudinal Employer-Household Dynamics Local Origin-Destination Employment Statistics.

Our specification is:

$$\Delta Est_{zc} = \beta_1 \Delta Pop_{zc} + \beta_2 WFH_Resi_{zc} + \beta_3 WFH_Emp_{zc} + X_{zc} + \gamma_c + \varepsilon_{zc}, \quad (1)$$

where ΔEst_{zc} is the percent growth in establishments in neighborhood z in city c , ΔPop_{zc} is the percent growth in customer residential population, and WFH_Resi_{zc} and WFH_Emp_{zc} measures pre-pandemic residential and employment exposure to WFH, respectively. We also include a vector of neighborhood characteristics, X_{zc} , which includes measures of pre-pandemic neighborhood non-white residential share and median income from the 2015-2019 5-year American Community Survey (ACS) data, a $Downtown_{zc}$ dummy to control for remaining differences in pandemic effects in downtown cores that are not captured

¹⁷Figure 10 shows how these exposure measures map to ZIPs along our 0 to 1 scaled distance within cities.

by our other controls, and the growth in online and offline residential spending among Chase customers in each neighborhood.¹⁸ In all specifications, we include city fixed effects, γ_c , and cluster standard errors by city.

3.2 Identification

Identifying the effects of WFH on establishment growth presents many challenges. For a causal interpretation of the β coefficients in equation 1, the COVID-19 induced adoption of WFH in this short two-year window needs to be an exogenous shifter of customer residential and employment locations that then fundamentally alters the viability of retail goods and service establishments in different locations. The clear break in parallel establishment growth within cities in Figure 8, even absent controls, supports this interpretation. The inclusion of neighborhood characteristics and spending growth by channel recognizes that the pandemic differentially affected the economic performance of low- and high-socioeconomic status neighborhoods and online shopping, which plausibly correlates with both WFH and establishment growth. These effects could operate through the direct impact of the disease and other indirect economic effects, such as through the labor market or stimulus payments (Chetty et al. 2022). These characteristics also control for which destination neighborhoods would have been more attractive to residents moving due to WFH and correlated merchant card payment adoption during the pandemic that could create more false entries in some neighborhoods. City fixed effects control for the common effects of the pandemic across each city. After controlling for these factors, we believe that reverse causality or omitted variable bias are not significant concerns in this context studying changes over a short two-year window. The limitation of this approach is that we cannot rule out that such an omitted variable exists.

Our empirical specification most closely follows Brueckner, Kahn and Lin (2023) in utilizing the exogeneity of pre-pandemic WFH exposure for identification. There are several important distinctions. First, they study the impact of residential moves on house prices with pre-pandemic residential exposure to WFH as an instrument for residential moves induced by WFH. In contrast, we control for residential moves and attribute those moves during our two-year window to the adoption of WFH. Thus, WFH residential exposure in our context is as an instrument for which neighborhoods were more likely to have residents who transition to WFH, conditional on residential population growth. Second, we build on their strategy by separately measuring the impact of WFH exposure coming from changes in workplace arrangements of the workers in a neighborhood as well as the residents.

¹⁸For each quarter q and the same quarter in 2019, we filter to customers (regardless of location) who make at least 10 transactions in each month of both quarters to focus on active customers. We aggregate the spending of these customers for each ZIP and quarter for the growth calculation. In this way, an active customer who moves from San Francisco to Atlanta in November of 2019 would have their October 2019 spending allocated to their San Francisco ZIP and their November and December 2019 spending to their Atlanta ZIP for Q4 2019.

3.3 Estimation results

Our results for aggregate establishment growth are found in Table 1. Echoing our descriptive figures, we first show in Model 1 that being in a downtown area is associated with an average 3.1 percentage point lower rate of growth in establishments, absent other controls. Model 2 adds our estimate of population growth for the same period. We find that the residential moves of people transitioning to WFH have a strong correlation with establishment growth during this time. The impact is substantial enough that the downtown dummy variable becomes insignificant. Adding instead our exposure measures in Model 3, only employment exposure has a statistically significant impact on establishment growth conditional on the downtown dummy. This points to the downtown dummy primarily capturing the effect of population loss due to WFH at the core, but not the employment effects. Once population loss is included in Model 4, the downtown dummy once again becomes insignificant. The stability of the coefficients across Models 2-4 suggests that the effects of WFH through population versus residential and employment exposure are relatively independent sources of variation in the data.

Table 1: Decomposition analysis of Q4 2019 - Q4 2021 establishment growth (%)

	1	2	3	4	5
Downtown Core	-3.108 (0.86)	-0.558 (0.811)	-2.203 (0.7)	-0.052 (0.624)	1.308 (0.84)
Population Growth (%)		0.354 (0.061)		0.329 (0.062)	0.318 (0.109)
WFH residential exposure (%)			-0.096 (0.071)	-0.063 (0.074)	-0.226 (0.108)
WFH employment exposure (%)			-0.135 (0.059)	-0.118 (0.053)	-0.106 (0.053)
Online spending growth (%)					0.046 (0.024)
Offline spending growth (%)					-0.102 (0.107)
Non-white residents (%)					-0.035 (0.018)
Median income (\$1000s)					0.037 (0.023)
Mean Est. growth	2.82	2.82	2.82	2.82	2.82
Observations	3653	3653	3653	3653	3653
R-sq.	0.038	0.060	0.048	0.066	0.072
Adj. R-sq.	0.034	0.055	0.043	0.061	0.066

Notes: All models include CBSA fixed effects. Standard errors in parentheses and clustered by city.

Sources: JPMorgan Chase, U.S. Census Bureau American Community Survey and Longitudinal Employer-Household Dynamics Local Origin-Destination Employment Statistics.

In our final specification in Model 5, we include spending by channel and other controls for neighborhood demographics. These controls quadruple the size of the negative coefficient for WFH residential exposure to -0.23 percentage points and make it statistically significant. This makes sense given that we

observe strong positive correlation among neighborhoods with more WFH residential exposure, higher income, and fewer non-white residents. The meaning of this effect is that, conditional on a population shock to a neighborhood due to WFH, a likely transition of residents in those locations to WFH reduces establishment growth. This matches our intuition that if consumers who WFH go on fewer trips overall, the impact of that daily consumption change will be large near their residential locations where most shopping trips originate. Beyond that, the impact of employment exposure is also substantial with a coefficient of -0.11. This points to the critical reliance of some retail on the commuters in employment locations.

A notable result from our main specification is the small and positive impact of online retail growth. This contrasts with the dominant narrative that online retail negatively affects physical stores (Dolfen et al. 2023; Chava et al. 2024). However, it's important to note that this measure reflects concurrent online retail growth. While online retail for goods continued to grow during the pandemic, the strongest growth was in online groceries and restaurants (Figure A.10). Since both product categories still depend on physical grocery stores and restaurants, this growth may have supported brick-and-mortar establishments, offsetting negative effects in more tradable goods. Additionally, spillover effects between online and physical retail across product categories can be positive (Relihan 2024). We leave further analysis of these online-offline retail interactions for future research.

Table 2 shows significant variation in the impact of WFH across establishment good and service product types. Starting with the largest product categories, grocery establishment growth is strongly associated with population growth but shows no significant correlation with WFH exposure through either residents or employment. This makes sense because most grocery shopping is tied to residential locations and residents are unlikely to lower their demand for food prepared at home when they work from home. In contrast, while restaurants are similarly responsive to residential population growth, there is a marginally statistically insignificant, but negative coefficient on WFH residential exposure. This lends support to substitution between food away from home and food produced at home. More striking is the strong negative association with WFH employment exposure. Indeed, restaurants are the online product category which drives the negative association between aggregate establishment growth and WFH employment exposure. For general goods, residential population growth has almost no independent association with establishment growth. Rather, for these establishments, the negative association with WFH residential exposure is the key driver for growth in this period, again suggesting substitution toward home production.

Outside of these largest products, and pharmacy (which shows little spatial variation in establishment growth), patterns in other product categories provide further insights. For clothing and personal care services, there is no statistically significant relationship with population and a negative effect for

WFH residential exposure. This is consistent with lower demand among residents who move in this period and where more residents transition to WFH. For home goods and services and leisure, there is a positive relationship with population growth and a statistically insignificant effect of WFH residential exposure. This suggests that these establishments are relocating with their customers, but are not otherwise significantly impacted by WFH. While the effect of population growth for professional consumer services is only marginally statistically insignificant, there is a significant and positive association with WFH residential exposure. This suggests that for these establishments, WFH is both causing these establishments to relocate and increasing demand for them.

3.4 Robustness

Time-span: One concern with our empirical analysis is that the two-year window is too short to learn about the long-run effects of WFH on retail markets. For instance, part of the disconnect for general goods between high suburban growth in transactions and declines in establishments may be the high capital costs of building new large general goods stores, which slows down their relocation with residential population. To understand how our empirical results change over time, we estimate our main specification for three additional two-year windows, with end dates of Q2 2020, Q4 2020, and Q2 2021. Table A.2 presents these results alongside our main result for the window ending in Q4 2021. While the impact of WFH home through population change and neighborhood exposure intensifies from Q2 2020 to Q2 2021 before moderating somewhat in Q4 2021, the impacts in each window are large and qualitatively similar. We also show in Figure A.2 that four-year growth rates for establishments from Q4 2019 to Q4 2023 across scaled distance to city center continue to show concave growth in which inner suburban establishment growth is substantially stronger than the downtown core or outer suburbs. Taken together, these results suggest that many establishments are not yet following new residential populations to the suburbs, though more relocation may occur over longer time horizons. Furthermore, while the point estimates for the effect of WFH exposure on retail establishment locations and composition may continue to moderate with more return-to-office, the economic significance and direction of those effects is likely robust.

Distance controls: In our main specification, we included a dummy for the downtown core as our control for distance to city center because Figure 8 shows the differences in population and establishment growth from 2019 Q4 to 2021 Q4 were most distinct between the downtown core and the rest of the city. Nonetheless, a continuous measure of distance to city center may serve as a better control for ZIP code features that vary over scaled distance. We report these results in Table A.1. The meaningful difference is that when using scaled distance, the coefficient on WFH residential exposure drops from -0.22 to

Table 2: Decomposition analysis of Q4 2019 - Q4 2021 establishment growth by product

	Grocery	General	Clothing	Rest.	Home	Personal	Leisure	Prof.	Pharmacy
Downtown Core	3.329 (1.343)	3.041 (1.77)	13.247 (6.786)	0.303 (1.518)	1.231 (2.258)	3.497 (1.306)	2.474 (4.464)	-0.411 (2.315)	1.271 (1.927)
Population Growth (%)	0.259 (0.149)	-0.069 (0.167)	0.383 (0.24)	0.31 (0.115)	0.235 (0.08)	0.089 (0.097)	0.367 (0.163)	0.205 (0.132)	-0.324 (0.172)
WFH residential exposure (%)	-0.219 (0.199)	-0.381 (0.16)	-0.684 (0.258)	-0.181 (0.119)	-0.121 (0.093)	-0.271 (0.087)	-0.37 (0.255)	0.203 (0.091)	-0.02 (0.135)
WFH employment exposure (%)	-0.031 (0.062)	0.087 (0.116)	0.115 (0.261)	-0.344 (0.062)	0.039 (0.078)	-0.037 (0.065)	-0.066 (0.113)	0.061 (0.076)	-0.023 (0.074)
Online spending growth (%)	0.001 (0.001)	0.003 (0.034)	0.03 (0.051)	-0.017 (0.018)	-0.007 (0.011)	-0.014 (0.007)	0.04 (0.059)	0.017 (0.012)	0.001 (0.005)
Offline spending growth (%)	0.116 (0.109)	0.077 (0.096)	0.204 (0.084)	0.036 (0.081)	0.078 (0.043)	0.174 (0.079)	-0.005 (0.016)	0.021 (0.024)	0.209 (0.068)
Non-white residents (%)	0.024 (0.029)	-0.037 (0.034)	0.031 (0.07)	0.012 (0.024)	0.075 (0.053)	-0.155 (0.036)	0.13 (0.037)	-0.09 (0.035)	-0.025 (0.024)
Median income (\$1000s)	0.164 (0.067)	0.147 (0.021)	0.035 (0.069)	0.078 (0.034)	0.027 (0.018)	0.026 (0.027)	0.098 (0.042)	-0.075 (0.016)	-0.013 (0.019)
Mean Est. growth	6.52	2.85	9.96	7.61	3.35	1.40	6.54	4.10	1.71
Observations	3513	3454	2842	3582	3300	3343	2979	3068	2888
R-sq.	0.043	0.026	0.032	0.062	0.018	0.095	0.030	0.012	0.026
Adj. R-sq.	0.037	0.020	0.024	0.056	0.012	0.089	0.022	0.005	0.019

Notes: All models include CBSA fixed effects. Standard errors in parentheses and clustered by city. Online and offline spending growth is with regard to the same product type of the dependent variable, rather than overall spending by channel.

Sources: JPMorgan Chase, U.S. Census Bureau American Community Survey and Longitudinal Employer-Household Dynamics Local Origin-Destination Employment Statistics.

-0.16, a decline of one quarter, and becomes statistically insignificant. In neither the case of using the downtown dummy or scaled distance as a control does distance have a statistically significant effect when population growth is included in the regression. The high correlation between population growth, scaled distance to city center, and WFH residential exposure means that we may need a larger sample size to detect an independent statistically significant effect of each given the control for population growth. Nonetheless, the qualitatively similar coefficient supports our conclusions.

4 Final Remarks

Our results are relevant for the firms in commercial real estate who must reassess property acquisitions, tenant strategies, and development plans as they adapt to shifting consumer demand driven by WFH. Over the short-run, the relocation of retail establishments can primarily operate through sorting among existing retail properties with higher vacancy rates in downtowns and lower vacancy rates in suburbs. The least capital intensive will be more mobile. Over time, the movement of people and establishments could encourage new developments that durably alter the location of retail. The consequent increase in amenities could further attract new residents, reinforcing the initial migration wave due to WFH.

The increasing suburbanization of retail will have ripple effects across local economies. Most immediately, cities may lose substantial sales and property tax revenue if new suburban residents spend less offline on products subject to sales tax and new establishments are outside city jurisdiction. Although retail employment is likely to strengthen in suburban areas, the shift might worsen existing disparities. For example, Black households, predominantly based in downtown areas and underrepresented in industries adaptable to remote work, may face increased spatial job mismatches and reduced access to retail amenities ([Miller 2023](#)).¹⁹

The relocation of retail establishments also has implications for aggregate economic growth by potentially reducing agglomeration benefits. With less consumption amenity advantage in large coastal cities and downtowns, there may be less reason for knowledge-intensive workers to jointly locate in specific neighborhoods. Furthermore, as discussed in [Atkin, Chen and Popov \(2022\)](#), establishments like coffee shops can provide valuable locations for meetings of workers in knowledge-intensive industries where face-to-face interactions are central to innovation. Such meetings may be less likely to take place if fewer retail establishments near employment centers can be sustained with only a partial return to office. These distributional and aggregate consequences underscore the importance of accounting for evolving patterns of work and consumption to fully understand the impact of WFH on cities and the appropriate public policy response.

¹⁹2019 shows only 8.8 percent of employees in the 2-digits NAICS industries (51, 52, 54, and 55) most exposed to WFH were Black, while 78 percent were White.

References

- Adams-Prassl, Abi, Teodora Boneva, Marta Golin, and Christopher Rauh.** 2020. “Inequality in the impact of the coronavirus shock: Evidence from real time surveys.” Journal of Public Economics, 189: 104245.
- Aguiar, Mark, and Erik Hurst.** 2007. “Life-Cycle Prices and Production.” American Economic Review, 97(5): 1533–1559.
- Aksoy, Cevat Giray, Jose Maria Barrero, Nicholas Bloom, Steven J. Davis, Mathias Dolls, and Pablo Zarate.** 2023. “Time Savings When Working from Home.” NBER Working Paper No. w30866.
- Almagro, Milena, and Tomás Domínguez-Iino.** 2022. “Location Sorting and Endogenous Amenities.” University of Chicago, Becker Friedman Institute for Economics Working Paper No. 2022-162.
- Althoff, Lukas, Fabian Eckert, Sharat Ganapati, and Conor Walsh.** 2022. “The Geography of Remote Work.” Regional Science and Urban Economics, 93: 103770.
- Atkin, David, Benjamin Faber, and Marco Gonzalez-Navarro.** 2018. “Retail Globalization and Household Welfare: Evidence from Mexico.” Journal of Political Economy, 126(1): 1–73.
- Atkin, David, M. Keith Chen, and Anton Popov.** 2022. “The Returns to Face-to-Face Interactions: Knowledge Spillovers in Silicon Valley.” NBER Working Paper No. w30147.
- Baker, Scott R., Robert A. Farrokhnia, Steffen Meyer, Michaela Pagel, and Constantine Yannelis.** 2020. “How Does Household Spending Respond to an Epidemic? Consumption During the 2020 COVID-19 Pandemic.” The Review of Asset Pricing Studies, 10(4): 834–862.
- Barnatchez, Keith, Leland D. Crane, and Ryan A. Decker.** 2017. “An Assessment of the National Establishment Time Series.” Board of Governors of the Federal Reserve System Finance and Economics Discussion Series 2017-10.
- Baum-Snow, Nathaniel.** 2020. “Urban Transport Expansions and Changes in the Spatial Structure of U.S. Cities: Implications for Productivity and Welfare.” Review of Economics and Statistics, 102(5): 929–945.
- Baum-Snow, Nathaniel, and Daniel Hartley.** 2020. “Accounting for Central Neighborhood Change, 1980–2010.” Journal of Urban Economics, 117: 103228.

- Benson, David, Serafin Grundl, and Richard Windle.** 2020. “How do Rural and Urban Retail Banking Customers Differ?” FEDS Notes. Washington: Board of Governors of the Federal Reserve System. <https://doi.org/10.17016/2380-7172.2513>. Accessed on 2024-03-18.
- Blevins, Jason R., Ahmed Khwaja, and Nathan Yang.** 2018. “Firm Expansion, Size Spillovers, and Market Dominance in Retail Chain Dynamics.” *Management Science*, 64(9): 4070–4093.
- Brueckner, Jan K., Matthew E. Kahn, and Gary C. Lin.** 2023. “A New Spatial Hedonic Equilibrium in the Emerging Work-from-Home Economy?” *American Economic Journal: Applied Economics*, 15(2): 285–319.
- Brynjolfsson, Erik, Yu Hu, and Mohammad S. Rahman.** 2009. “Battle of the Retail Channels: How Product Selection and Geography Drive Cross-Channel Competition.” *Management Science*, 55(10): 1755–1765.
- Chava, Sudheer, Alexander Oettl, Manpreet Singh, and Linghang Zeng.** 2024. “Creative Destruction? Impact of E-Commerce on the Retail Sector.” *Management Science*, 70(4): 2168–2187.
- Chetty, Raj, John N. Friedman, Nathaniel Hendren, and Michael Stepner.** 2022. “The Economic Impacts of COVID-19: Evidence from a New Public Database Built Using Private Sector Data.” NBER Working Paper No. w27431.
- Cleveland, William S.** 1979. “Robust Locally Weighted Regression and Smoothing Scatterplots.” *Journal of the American Statistical Association*, 74(368): 829–836.
- Couture, Victor, and Jessie Handbury.** 2020. “Urban Revival in America.” *Journal of Urban Economics*, 119: 103267.
- Couture, Victor, Cecile Gaubert, Jessie Handbury, and Erik Hurst.** 2021a. “Income Growth and the Distributional Effects of Urban Spatial Sorting.” *Review of Economic Studies*, 3(1).
- Couture, Victor, Jonathan I. Dingel, Allison E. Green, Jessie Handbury, and Kevin R Williams.** 2021b. “Measuring Movement and Social Contact With Smartphone Data: A Real-Time Application to COVID-19.” *Journal of Urban Economics*, 127: 103328.
- Crane, Leland D., Ryan A. Decker, Aaron Flaaen, Adrian Hamins-Puertolas, and Christopher Kurz.** 2022. “Business Exit During the COVID-19 Pandemic: Non-traditional Measures in Historical Context.” *Journal of Macroeconomics*, 72: 103419.
- Dalton, Michael, and Jeffrey A. Groen.** 2022. “Telework During the COVID-19 Pandemic.” *Monthly Labor Review*, 1–15.

- Davis, Donald R, Jonathan I. Dingel, Joan Monras, and Eduardo Morales.** 2019. “How Segregated is Urban Consumption?” Journal of Political Economy, 127(4): 1684–1738.
- Davis, Morris A., Andra C. Ghent, and Jesse Gregory.** 2024. “The Work-From-Home Technology Boon and its Consequences.” The Review of Economic Studies, rda114.
- Davis, Peter.** 2006. “Spatial Competition in Retail Markets: Movie Theaters.” The RAND Journal of Economics, 37(4): 964–982.
- Decker, Ryan A., and John Haltiwanger.** 2022. “Surging Business Formation in the Pandemic: Causes and Consequences.” [Link](#). Accessed on 2023-06-02.
- De Fraja, Gianni, Jesse Matheson, Paul Mizen, James Rockey, and Shivani Taneja.** 2022. “Remote Working and the New Geography of Local Service Spending.” [Link](#). Accessed on 2023-08-24.
- Delventhal, Matthew J., and Andrii Parkhomenko.** 2023. “Spatial Implications of Telecommuting.” [Link](#). Accessed on 2023-07-14.
- Delventhal, Matthew J., Eunjee Kwon, and Andrii Parkhomenko.** 2022. “JUE Insight: How do cities change when we work from home?” Journal of Urban Economics, 127: 103331. JUE Insights: COVID-19 and Cities.
- Diamond, Rebecca.** 2016. “The Determinants and Welfare Implications of US Workers’ Diverging Location Choices by Skill: 1980-2000.” American Economic Review, 106(3): 479–524.
- Dingel, Jonathan I., and Brent Neiman.** 2020. “How Many Jobs Can be Done at Home?” Journal of Public Economics, 189: 104235.
- Ding, Lei, and Jackelyn Hwang.** 2022. “Has COVID Reversed Gentrification in Major US Cities? An Empirical Examination of Residential Mobility in Gentrifying Neighborhoods During the COVID-19 Crisis.” FRB of Philadelphia Working Paper No. 22-20.
- Dolfen, Paul, Liran Einav, Peter J Klenow, Benjamin Klopach, Larry Levin, and Wayne Best.** 2023. “Assessing the Gains from E-Commerce.” American Economics Journal: Macroeconomics, 15(1): 342–370.
- Duranton, Gilles, and Jessie Handbury.** 2023. “Covid and Cities, Thus Far.” [Link](#). Accessed on 2023-02-01.
- Ellickson, Paul B., Paul L.E. Grieco, and Oleksii Khvastunov.** 2020. “Measuring competition in spatial retail.” The RAND Journal of Economics, 51(1): 189–232.

- Fairlie, Robert.** 2020. “The Impact of COVID-19 on Small Business Owners: Evidence from the First Three Months After Widespread Social-Distancing Restrictions.” Journal of Economics & Management Strategy, 29(4): 727–740.
- Ferreira, Fernando V., and Maisy Wong.** 2022. “Neighborhood Choice After COVID: The Role of Rents, Amenities, and Work-From-Home.” NBER Working Paper No. w29960.
- Forsythe, Eliza, Lisa B. Kahn, Fabian Lange, and David Wiczer.** 2020. “Labor demand in the time of COVID-19: Evidence from vacancy postings and UI claims.” Journal of Public Economics, 189: 104238.
- Ganong, Peter, and Pascal Noel.** 2019. “Consumer Spending during Unemployment: Positive and Normative Implications.” American Economics Review, 109(7): 2383–2424.
- Glaeser, Edward L., Jed Kolko, and Albert Saiz.** 2001. “Consumer City.” Journal of Economic Geography, 1(1): 27–50.
- Goolsbee, Austan, and Chad Syverson.** 2021. “Fear, lockdown, and diversion: Comparing drivers of pandemic economic decline 2020.” Journal of Public Economics, 193: 104311.
- Gorback, Caitlin.** 2022. “Ridesharing and the Redistribution of Economic Activity.” [Link](#). Accessed on 2022-12-04.
- Gupta, Arpit, Vrinda Mittal, and Stijn Van Nieuwerburgh.** 2022. “Work From Home and the Office Real Estate Apocalypse.” NBER Working Paper No. 30526.
- Haltiwanger, John, Ron S. Jarmin, and Javier Miranda.** 2013. “Who Creates Jobs? Small versus Large versus Young.” The Review of Economics and Statistics, 95(2): 347–361.
- Holian, Matthew J.** 2019. “Where is the City’s Center? Five Measures of Central Location.” Cityscape, 21(2): 213–226.
- Hotelling, Harold.** 1929. “Stability in Competition.” The Economic Journal, 39(153): 41–57.
- Hsieh, Chang-Tai, and Esteban Rossi-Hansberg.** 2023. “The Industrial Revolution in Services.” Journal of Political Economy Macroeconomics, 1(1): 3–42.
- Jaravel, Xavier, and Martin O’Connell.** 2020. “Real-time price indices: Inflation spike and falling product variety during the Great Lockdown.” Journal of Public Economics, 191: 104270.
- Jia, Panle.** 2008. “What Happens When Wal-Mart Comes to Town: An Empirical Analysis of the Discount Retailing Industry.” Econometrica, 76(6): 1263–1316.

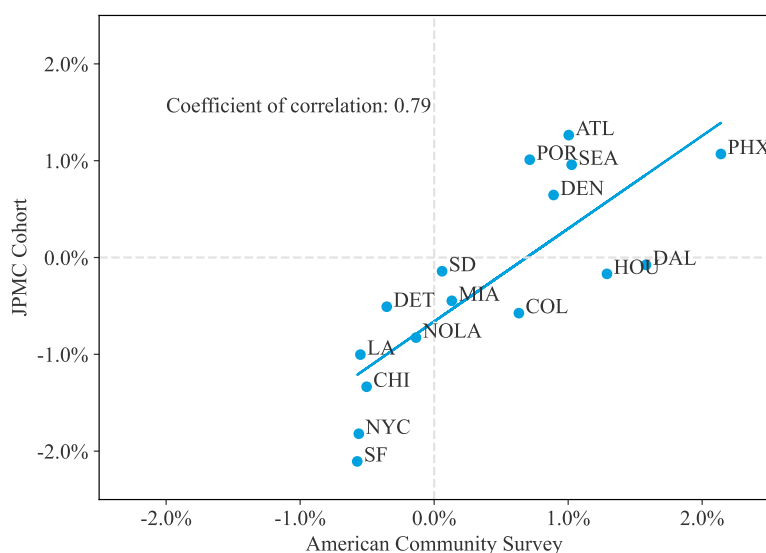
- Klopack, Ben, and Fernando Luco.** 2024. "Measuring local consumption with payment cards and cell phone pings."
- Klopack, Ben, Eric Lewis, and Fernando Luco.** 2024. "Economic Consequences of Natural Disasters: The Impact of Hurricane Harvey on Local Retail and Consumer Welfare." [Link](#). Accessed on 2024-09-23.
- Li, Wenli, and Yichen Su.** 2021. "The Great Reshuffle: Residential Sorting During the COVID-19 Pandemic and Its Welfare Implications." [Link](#). Accessed on 2022-12-01.
- Miller, Conrad.** 2023. "When Work Moves: Job Suburbanization and Black Employment." NBER Working Paper No. 24728.
- Miyauchi, Yuhei, Kentaro Nakajima, and Stephen J. Redding.** 2022. "The Economics of Spatial Mobility: Theory and Evidence Using Smartphone Data." [Link](#). Accessed on 2022-12-04.
- Monte, Ferdinando, Charly Porcher, and Esteban Rossi-Hansberg.** 2023. "Remote Work and City Structure." [Link](#). Accessed on 2023-04-30.
- Oh, Ryungha, and Jaeun Seo.** 2023. "What Causes Agglomeration of Services? Theory and Evidence from Seoul." [Link](#). Accessed on 2024-1-30.
- Ramani, Arjun, and Nicholas Bloom.** 2021. "The Donut Effect of COVID-19 on Cities." NBER Working Paper No. w28876.
- Relihan, Lindsay.** 2024. "Clicks and Bricks: The Complementarity of Online Retail and Urban Services." [Link](#). Accessed on 2024-1-30.
- Relihan, Lindsay, Marvin Ward, Chris Wheat, and Diana Farrell.** 2020. "The Early Impact of COVID-19 on Local Commerce: Changes in Spend Across Neighborhoods and Online." Covid Economics, 1–28.
- Rosenthal, Stuart S., William C. Strange, and Joaquin A. Urrego.** 2021. "Are City Centers Losing Their Appeal? Commercial Real Estate, Urban Spatial Structure, and COVID-19." Journal of Urban Economics, 127: 103381.
- Seim, Katja.** 2006. "An empirical model of firm entry with endogenous product-type choices." The RAND Journal of Economics, 37(3): 619–640.
- Van Nieuwerburgh, Stijn.** 2022. "The Remote Work Revolution: Impact on Real Estate Values and the Urban Environment." Real Estate Economics.

5 Online Appendix

5.1 Measuring population

We measure the location and activity of an individual person using their accounts with the bank across potentially multiple products, including deposit, credit, and mortgage products. The bank-provided identifiers that attempt to link product accounts together when they appear to belong to the same person are a sometimes imperfect proxy for a single individual, as with shared accounts. Where credit and debit card accounts are associated with multiple people, we assign all transactions to the primary account holder and use the location information associated with the primary account holder. We would then count the change in location for the primary account holder as a residential move of one person, even though an entire household may move. Other entirely unobserved members of a household, such as children, are not measured in our data.

Figure A.1: External population growth comparison



Notes: This figure shows population growth in the JPMCI cohort from Q3 2019 to Q3 2020 relative to the 2019 - 2020 Census estimates for each city.

Sources: JPMorgan Chase, U.S Census estimates from Table B of [Frey \(2022\)](#).

Figure A.1 shows how the JPMCI cohort for population change estimates from Q3 2019 to Q3 2020 by city compares against the latest Census estimates, as calculated by [Frey \(2022\)](#). The one-year changes in population by city closely match the Census estimates without any sample weighting of the JPMCI cohort. This is by design, as we focus on cities where the bank has the largest footprint and serves a broad cross-section of the population as customers. Of course, there is generally less representation at the lower end of the socioeconomic spectrum where the population is more likely to be underbanked. To the extent that the coverage of different populations vary across space, that can contribute to some error. Finally, note that, on average, the JPMCI cohort suggests slightly less population growth than the

Census. Besides differences in population coverage, this could also reflect delays in updating addresses reported to the bank over the one-year horizon.

5.2 Product mapping

The basis for our product categorization are Merchant Category Codes (MCCs) defined by the International Organization for Standards. The codes' purpose is to define the primary good or service the firm is engaged in and are used to detect fraud, allocate reward points, and determine tax and interchange rates for different purchases. They are typically assigned at firm birth by a credit card network (or when card terminals are first adopted by the firm). Multiple codes may be assigned to different lines of business at the same firm and different credit card networks may assign different MCCs to the same firm. MCCs are fairly static over time so that even if a firm's primary good or service changes, the MCC would not. For example, a firm that starts by selling books but goes on to sell a wide-variety of general merchandise may continue to be classified as a book store. Some firms, like airlines, are assigned their own MCC.

We map MCCs to a broader product type based on our analysis of the merchants that commonly appear in each MCC in our data. We believe our custom mapping to be more appropriate than other mappings for data categorized by MCC, which first map between MCCs and 6-digit North American Industry Classification System (NAICS) codes based only on similar code descriptions and then aggregates to 3-digit or 2-digit NAICS. Issues arise with the latter in this context because even though it is intuitive that the vast majority of our card transactions reflect retail spending on goods and services, relying only on similar descriptions of 6-digit NAICS codes would lead many of our transactions to be categorized as manufacturing, wholesale trade, and construction when this is clearly not reflective of the actual spending. For example, one may find it natural to map MCC 5462: Bakeries to the 6-digit NAICS code 311811: Retail Bakeries. However, this is part of the 2-digit NAICS code: Manufacturing. We find this inappropriate as the typical bakery we observe in our data is a small retail outlet rather than a large manufacturing operation. It would be more appropriate to map to the 6-digit NAICS 445291: Baked Goods Retailers.

We do not, however, strictly map MCCs to particular 2-, 3-, 4-, or 6- digit NAICS codes for two reasons. First, NAICS codes are better optimized to classify manufacturing-related industries rather than retail-related goods and services. We find it more natural to group some goods and services together in the same product category because of their closely related consumption. For example, video game rentals, online streaming services, books and golf courses are a selection of goods and services related to leisure activities that are well-represented by card spending. These would map to different 2-digits NAICS including 44-45: Retail Trade, 51: Information, and 71: Arts, Entertainment, and Recreation, which would obscure patterns in everyday leisure-related consumption if aggregated separately. Fur-

thermore, broad-swaths of the latter 2-digit NAICS are not well-represented by card spending.

Second, in observing the firms that appear in particular MCC or NAICS codes, there is often no clear one-to-one correspondence. For example, home internet providers and video streaming services appear in the same 6-digit NAICS 516210, but separate MCC codes which then themselves include firms in other 6-digit NAICS. These issues particularly relate to our leisure goods and services, professional consumer services, home goods and services, and general goods product categories. However, for the remaining products, there is a more clear (though not strict) cross-walk. Clothing maps to 4-digit NAICS 4481 and 4482; Restaurant to 3-digit NAICS 722; Grocery to 3-digit NAICS 445; Personal Care Goods and Services to 4-digit NAICS 8121 and 8123; and Pharmacy to 6-digit NAICS 456110.

Finally, we complement our MCC mapping with custom mapping for large online-only retailers in grocery, restaurant, and general goods that tend to fall into residual MCC codes that capture miscellaneous spending not elsewhere categorized. To do this, we use regular expressions to find merchant transaction descriptions which include the names of these retailers.

5.3 Measurement extensions

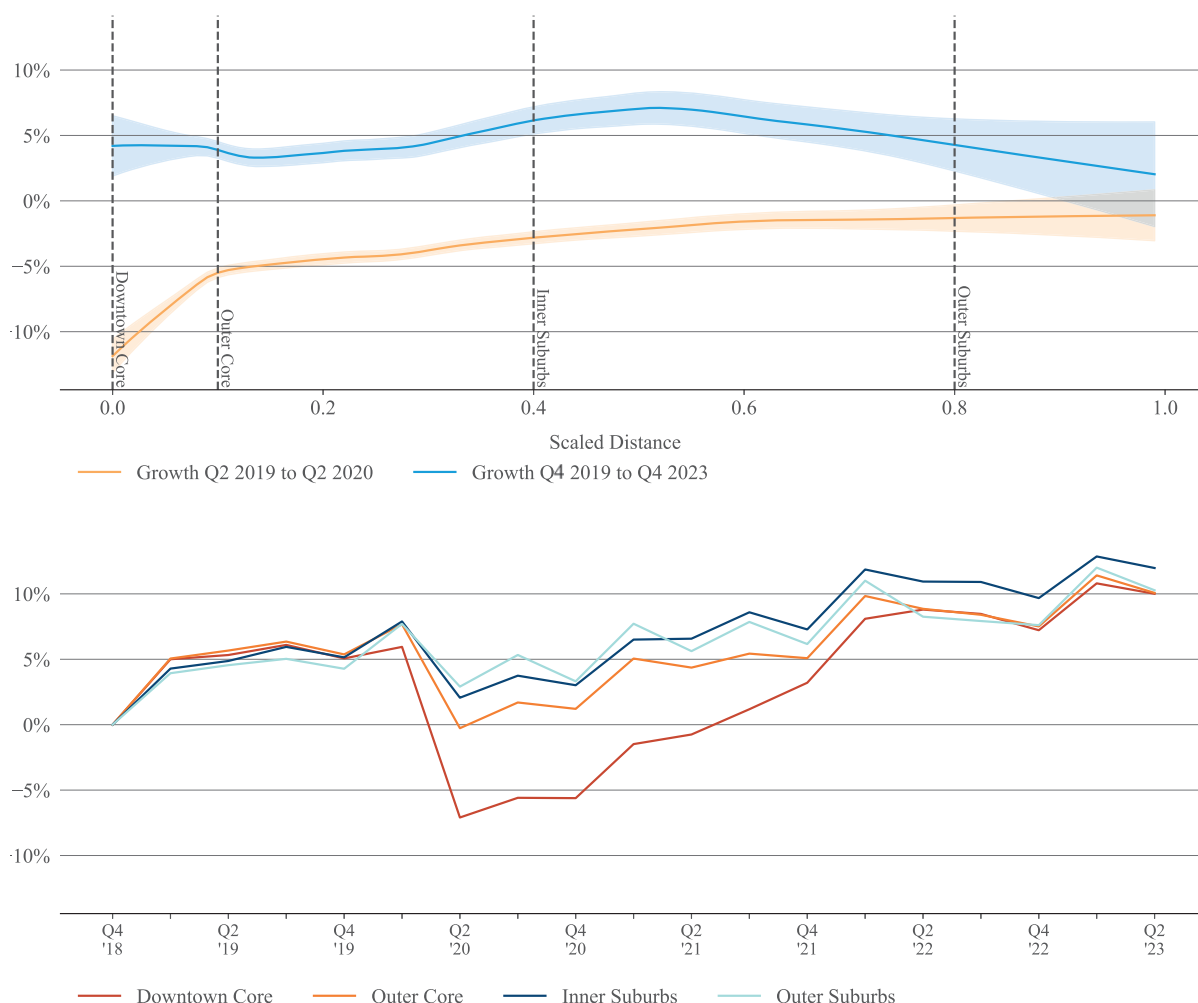
Our main analysis focuses on the measurement of the extensive margin of establishment entry and exit from Q4 2019 to Q4 2021, differentiating establishments only by their location and product. In this section, we detail straightforward extensions of establishment measurement with our data.

Figure A.2 shows the most recent data for within city establishment growth. As of Q2 2023, all locations within cities show positive growth relative to the either 2018 or 2019 as a base year. However, the inner-suburban growth continues to dominate.²⁰

Differentiation of establishments by size is a desirable feature of any measure used to evaluate establishment performance. We classify establishment size based on the presence of a unique establishment number, which is almost always included in transaction descriptions at terminals for chain-operated retail establishments. The distinction of chain versus independent establishment growth is important for retail markets, as this is one of the most important margins identified in the literature in explaining retail dynamics (Blevins and Yang 2018; Kleinman 2023). Figure A.3 illustrates growth, along with entry and exit rates, based on this size classification. Our findings indicate that, although smaller independent establishments faced more significant challenges at the pandemic's outset, they rebounded more swiftly, returning to pre-pandemic entry rates faster than their larger chain counterparts. Additionally, we observe that, before the pandemic, chain establishments had lower entry and exit rates compared to independent ones, aligning with rates reported by the BDS and QCEW. This pattern lends additional credibility to

²⁰We can provide similar population updates pending revision in our data assets of newer bank-provided identifiers used for customers.

Figure A.2: Extended establishment growth within cities



Notes: Scaled distance is calculated as the distance between the centroid of each ZIP and the city hall of the principal city of its CBSA, then dividing this distance by the greatest ZIP to city hall distance in the CBSA. In panel (a), we index growth for both population and establishment growth for the early (Q2 2020) and late (Q4 2023) to the corresponding quarter in 2019 to abstract from seasonality differences in growth. Bootstrapped 95 % confidence interval bands are shown as shaded areas in the top panel. In panel (b), growth in each quarter is relative to the same quarter in 2018.

Sources: JPMorgan Chase.

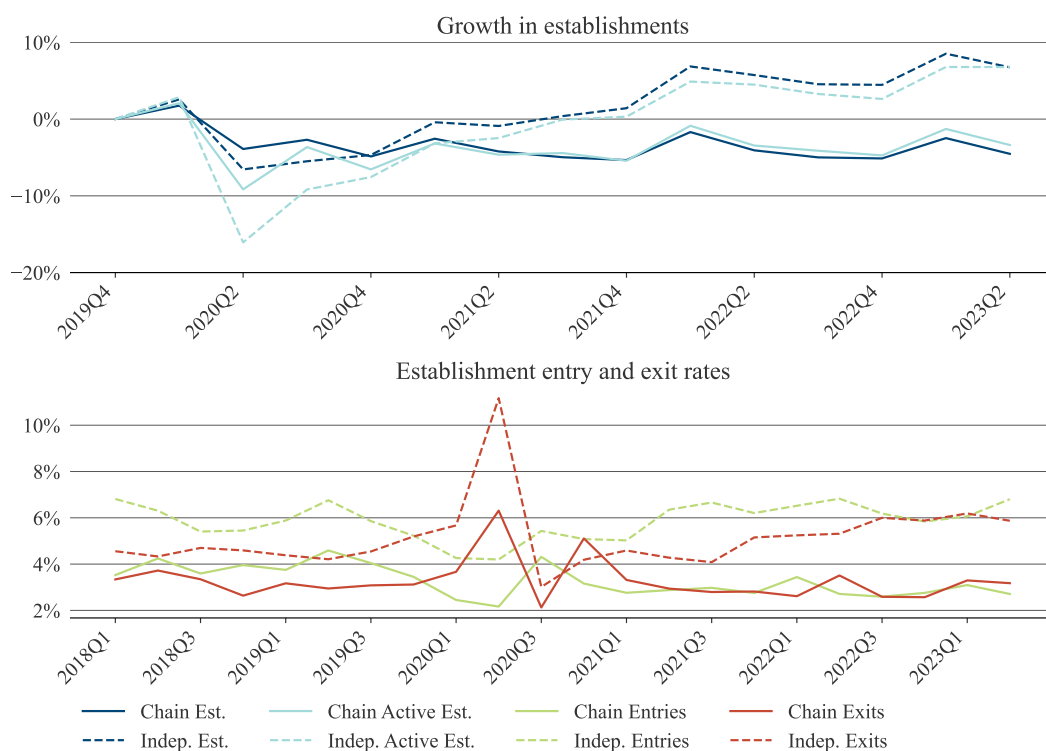
our method's ability to capture a broader spectrum of small, non-employer establishments.²¹

Figure A.4 shows spending at establishments by product within cities. We note that intensive margins, including transactions volume and spending, are generally more subject to Chase customer base effects.²² For the sixteen cities included in this analysis with a geographically stable customer base during our sample period, we can report these with minimal impact for Chase customer base changes because pandemic effects certainly dominate marginal customer base changes and most population movement is within cities. For other cities and time periods, measurement of establishment-specific transaction vol-

²¹Our card transaction-based data do not allow for classifying establishments by workforce size directly.

²²For example, a new bank branch in a neighborhood could attract new customers who enter our data, possibly generating additional transactions in our view without any real economic increase in trips or spending. For this reason, a best practice with our data is to create a stable customer base for spending analysis, as we do in the appendix Section 5.5.

Figure A.3: Chain and Independent

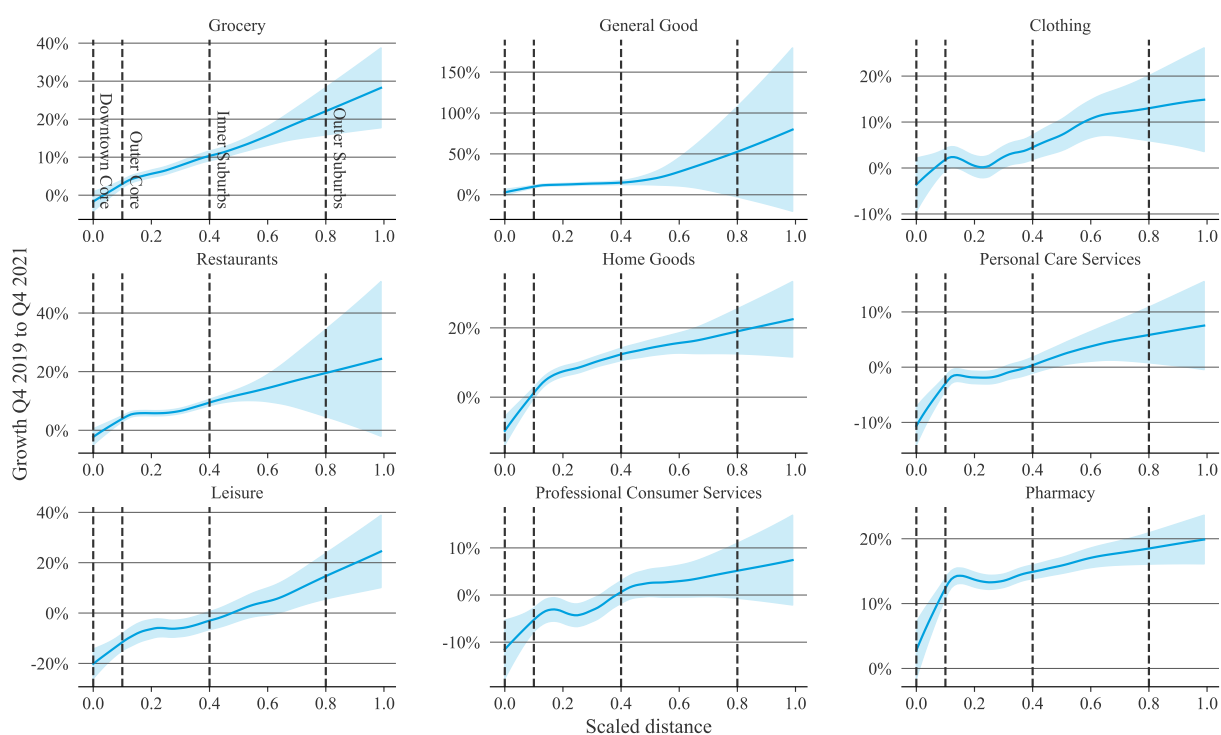


Notes: The top panel shows growth in establishments versus active establishments for chain versus independent retailers. Growth is relative to the same quarter in 2019. Active establishments are those at which customers make transactions in the quarter. The bottom panel shows the entry and exit rate relative to the stock of establishments in the previous quarter.

Sources: JPMorgan Chase.

umes and spending should be made from a stable customer base. Future versions of our establishment measure can link each establishment back to the individual customer transactions to create cross-time stable customers bases for such a measurement on an ongoing basis.

Figure A.4: Spending growth at establishments within cities by product



Notes: Growth is for Q4 2019 to Q4 2021. Scaled distance is calculated as the distance between the centroid of each ZIP and the city hall of the principal city of its CBSA, then dividing this distance by the greatest ZIP to city hall distance in the CBSA.

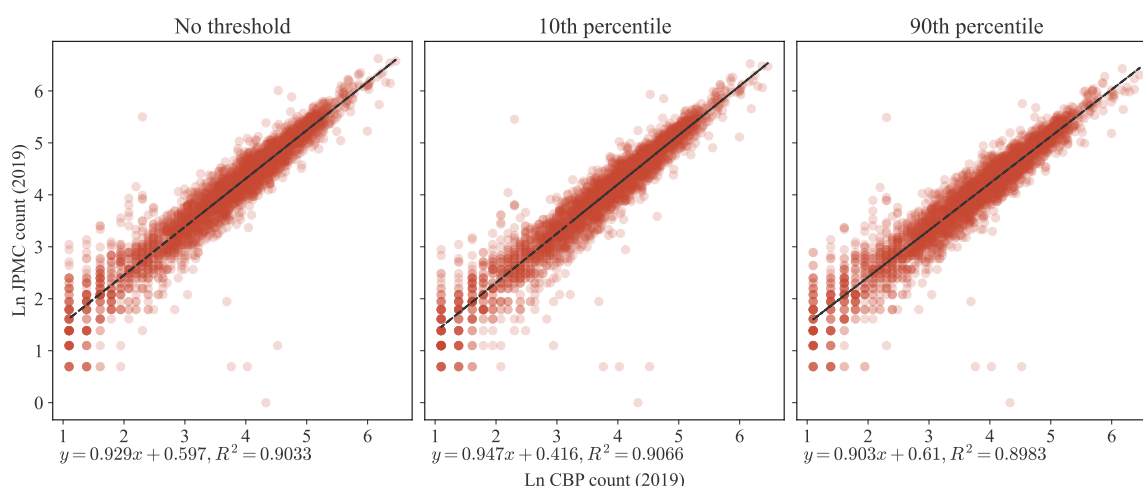
Sources: JPMorgan Chase.

5.4 Further comparing establishments to the CBP

An important factor likely to drive overcounting in JPMCI relative to the CBP is coverage by JPMCI of small locations that are sole proprietorships or where few people work. This would reflect the same source of discrepancy that is well-documented between Census' payroll-based datasets and alternate private sources of establishment information, as surveyed in [Barnatchez, Crane and Decker \(2017\)](#). While we cannot test this directly in our data because we lack a view on establishment employment, this issue is less likely to affect restaurants, as most have employees; indeed, this is where we overcount the least.

We can, however, test whether these differences are due to coverage at the bottom of the establishment distribution. In Figure A.5 we show how our match in counts to the CBP for restaurants in 2019 changes when we remove the bottom ten and top ten percent of restaurants ranked by their transaction activity.²³ The results show that removing the bottom 10 percent of restaurants reduces our count of restaurants in ZIP codes where the CBP measures few restaurants, pushing the intercept toward zero. Removing these restaurants also pushes the slope toward one and slightly improves the linear fit. In contrast, removing the top ten percent of restaurants creates the opposite effect by removing restaurants in zips where both our measure and the CBP agree that there are many restaurants. Thus, we are not simply overcounting everywhere at the same rate relative to the CBP, but particularly where our measure detects more very small establishments.

Figure A.5: Comparison excluding small and large restaurants

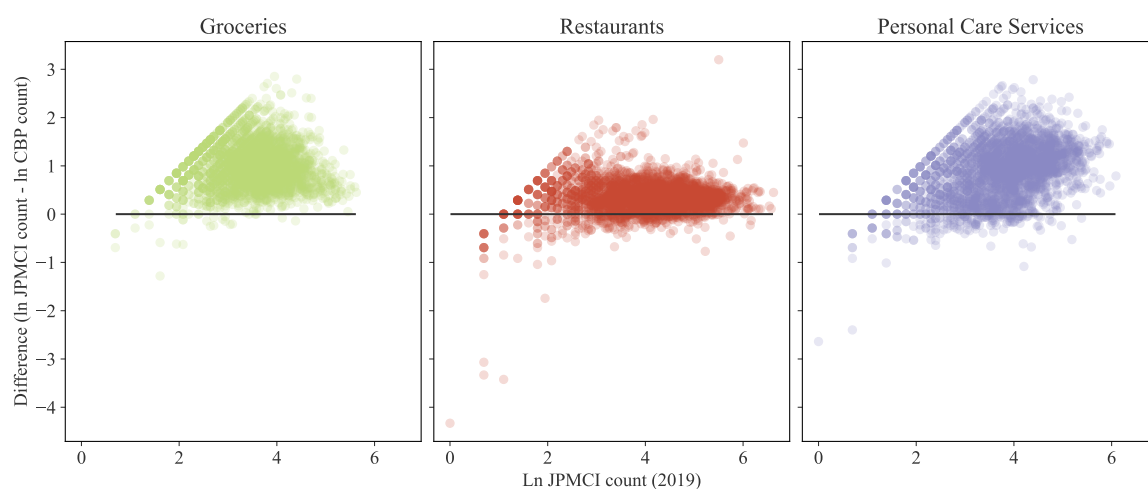


Notes: The left panel of this figure shows the correspondence in establishment counts by ZIP code for restaurants in the JPMCI measure for Q1 2019 with the CBP as collected in March 2019. The middle and right panel show the correspondence when establishments above and below the tenth and ninetieth percentile of transaction volume, respectively, are removed from the JPMCI measure. Percentiles are calculated using only restaurants and are specific to each city.

Sources: JPMorgan Chase, U.S. Census Bureau County Business Patterns.

²³Transaction ranking is within product and city to account for different baseline transaction activity across products and Chase customer coverage across cities, respectively. For example, the bottom ten percent of restaurants in New York City have fewer than 16 card transactions in Q1 2019.

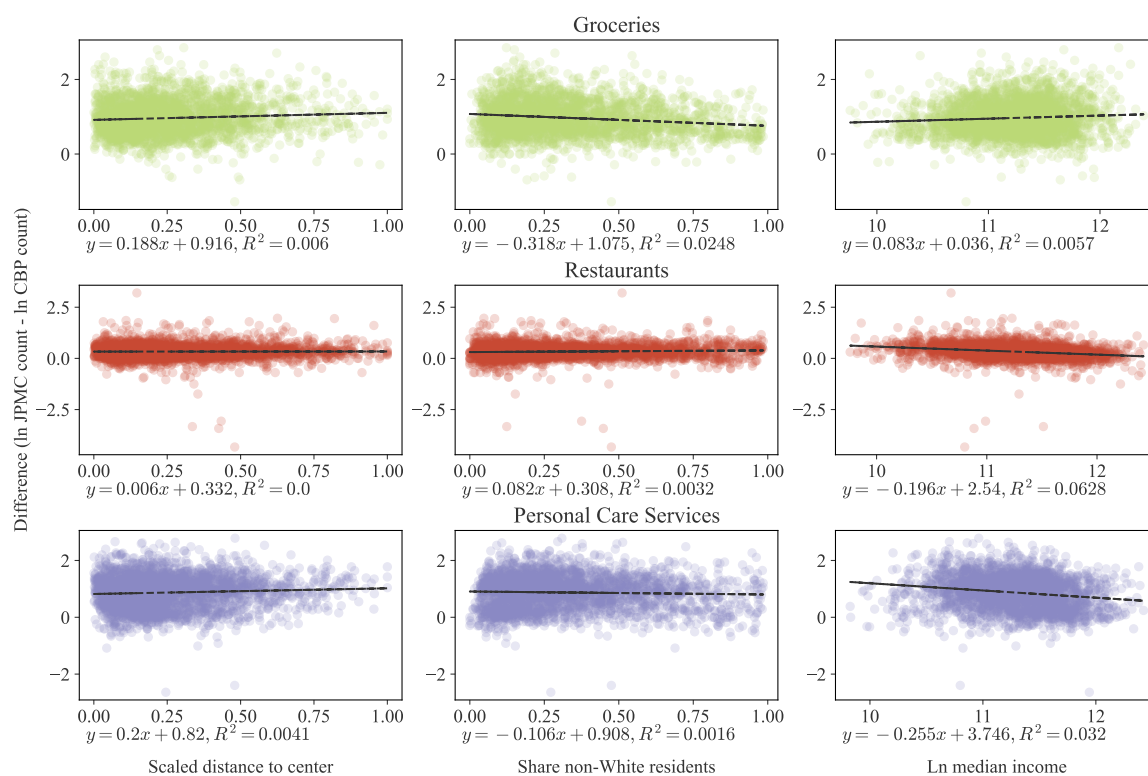
Figure A.6: Difference with CBP by count of retail establishments



Notes: This figure shows differences in establishment counts by ZIP code between the JPMCI and CBP measures by the count of the establishments measured by JPMCI.

Sources: JPMorgan Chase, U.S. Census Bureau County Business Patterns.

Figure A.7: Difference with CBP by ZIP characteristics



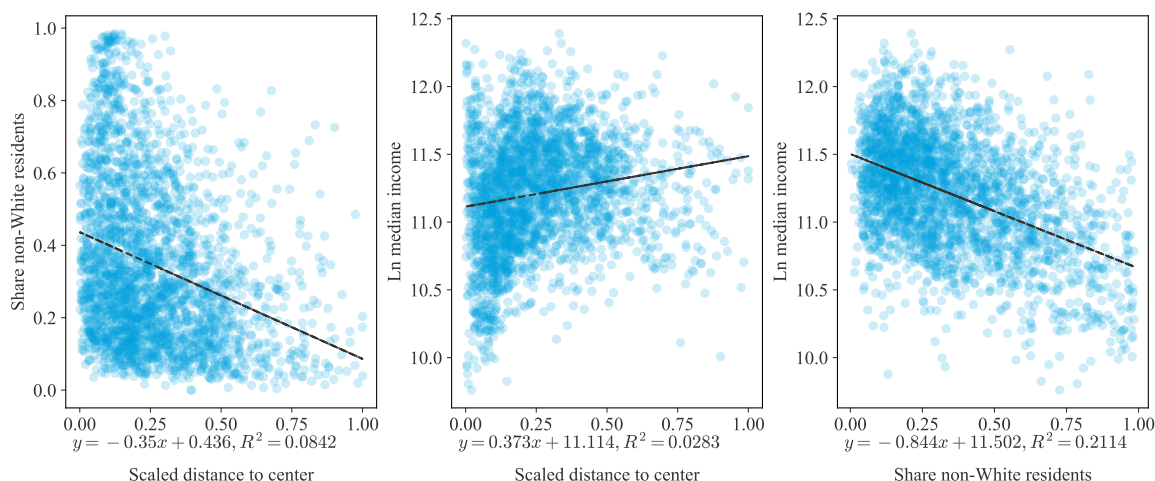
Notes: This figure shows differences in establishment counts by ZIP code between the JPMCI and CBP measures across ZIP characteristics from Census data of distance to center, non-white residents, and median income.

Sources: JPMorgan Chase, U.S. Census Bureau County Business Patterns and American Community Survey.

We next focus on understanding the variation in differences between the JPMCI and CBP counts observed around the trend line. To do this, we first plot the difference in counts by the JPMCI count of establishments in 2019. This is shown in Figure A.6. The first feature to call attention to with this view is that the JPMCI and CBP measures more closely agree on where there are few or many of each estab-

lishment type. Where they primarily differ is in locations where the CBP measures a moderate number of each establishment and JPMCI measures many more, like with very small restaurants. Secondly, this view makes clearer that while the JPMCI measure generally overcounts relative to CBP, JPMCI undercounts relative to CBP for some zip codes. These would be all the points that lie below the horizontal bar in each panel. Undercounting is more likely for restaurants and personal care services, products that likely include more unobserved all-cash establishments than groceries.

Figure A.8: Correlation in ZIP characteristics



Notes: This figure shows two-way correlations between the three ZIP characteristics, distance to center, non-white residents, and median income, for ZIPS used in the analysis.

Sources: JPMorgan Chase, U.S. Census Bureau American Community Survey.

In Figure A.7 we plot these difference across ZIP code features, including distance to center, share non-white residents, and median income. The latter two features are pulled from the same ACS data used in our main results. For context, we also show the two-way scatter plots of each pair of ZIP codes features in Figure A.8. There are a number of notable patterns that emerge:

1. The ZIP codes with the most extreme negative differences tend to occur in the middle of the scaled distance of cities. Inspection of the small number of these specific zip codes reveals that they tend to occur in the suburban areas of a few CBSAs where there is significant presence of non-Chase bank branches and, thus, likely weaker coverage of local residents. While this is a minor issue in this analysis in cities where Chase is a dominant presence, this highlights the necessity of good resident coverage for measurement of retail establishments with card transactions.
2. For restaurants and personal care services there are, a some ZIP codes with extreme positive differences in the suburbs. Inspection reveals that they tend to occur in locations with unusual features. For example include specialty leisure offerings, such as ski resorts, beach towns, and casinos. We measure them as collections of different establishments via their different transaction descriptions, but may be considered as one establishment when measured from an employment-

based view. These types of locations tend to also attract many temporary establishments, like refreshment stands or vending machines, which may similarly be reflected in transactions data, but not other sources. We also observe colleges and universities with dining halls and gyms in some ZIP codes, while education establishments are specifically excluded from the CBP.

3. The JPMCI measure overcounts relative to CBP in more low-income neighborhoods for restaurants and personal care services. This makes sense if low-income neighborhoods have higher concentrations of small or non-employer establishments for these product types.
4. In contrast, the observed pattern for grocery stores strongly suggests a “farmers market effect”. For grocery stores we observe more overcounting relative to CBP where there are more white residents and higher median income, a pattern which is not present for the other products. Inspection of specific ZIP codes is consistent with the interpretation that we pick up many temporary and small establishments that sell food that may not be recorded as employment locations by firms.

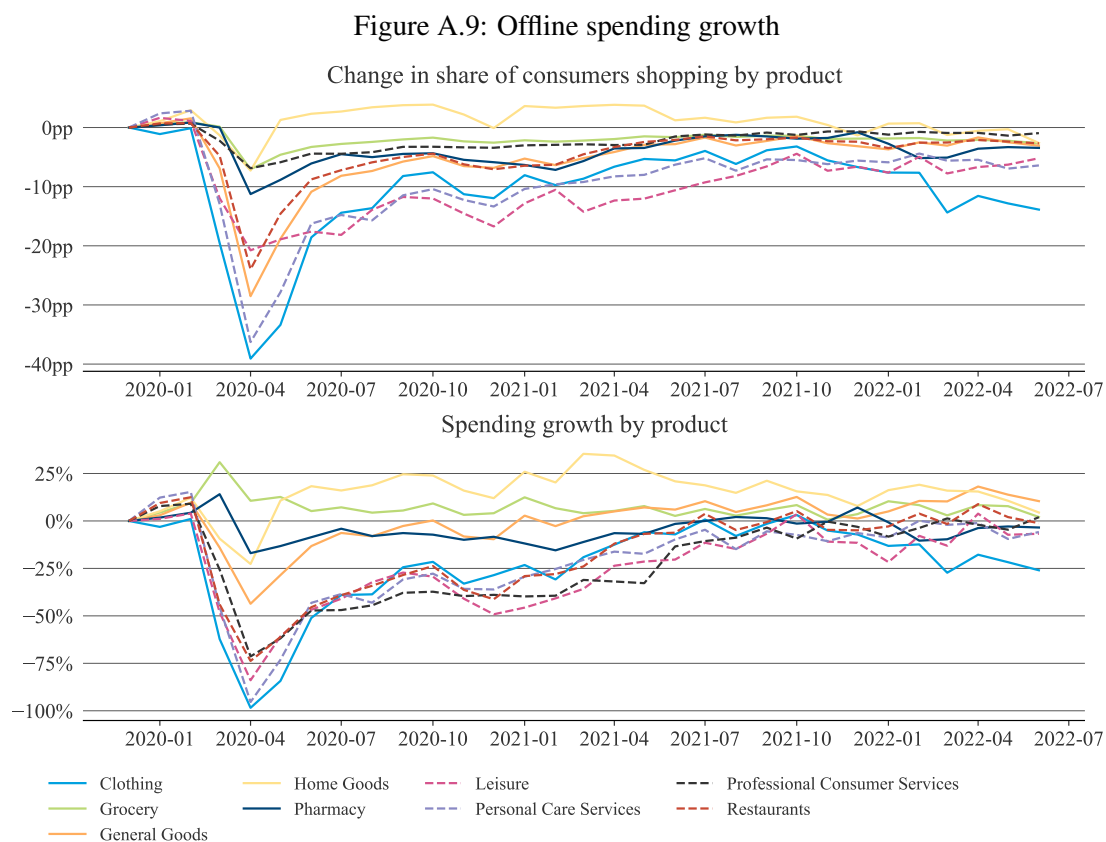
5.5 Changes in spending across channel

In this section, we investigate the persistence and scale of spending growth across products and channels during the pandemic. Doing this for offline spending is important because the validity of measuring brick-and-mortar establishments via offline transactions during our sample period rests in part on the sufficiency of offline transactions to distinguish true exits and entries from inactivity related directly to the pandemic. For example, continued lockdowns or fears of contagion could prevent a return to offline purchasing activity during our sample period, creating false exits that are continued inactivity. Any substantive, persistent changes in online spending behavior can also inform possible effects of the pandemic that work in concert with WFH.

To investigate these features, we construct a panel comprising 10.6 million individuals in any location who made at least 10 transactions per month from July 2018 to June 2022. For online products, we limit our focus to the five largest product categories (as measured by total spending across channel). For each product and channel, we analyze both the percentage of people using the products in the channel each month and the total spending on each product in the channel. This is itself an important data contribution, as tracking online growth by product is not currently possible with public data sources and these differences across product are distinct during the pandemic.

Figure A.9 provides the results for offline spending. Most critically for the main results of this paper, we see in the top panel that by Q4 2021 the frequency of customers shopping for each product type offline each month is within ten percentage points of the pre-pandemic baseline. This suggests that pandemic forces preventing a return to offline spending were largely mitigated by that time. Therefore, there is minimal loss of consumers whose transactions can survey establishments in the final quarters

we use to measure establishment activity. Total spending recovered even further by Q4 2021, though this can reflect both an increase in customer trips and other other spending changes, such as the amount purchased and product prices.



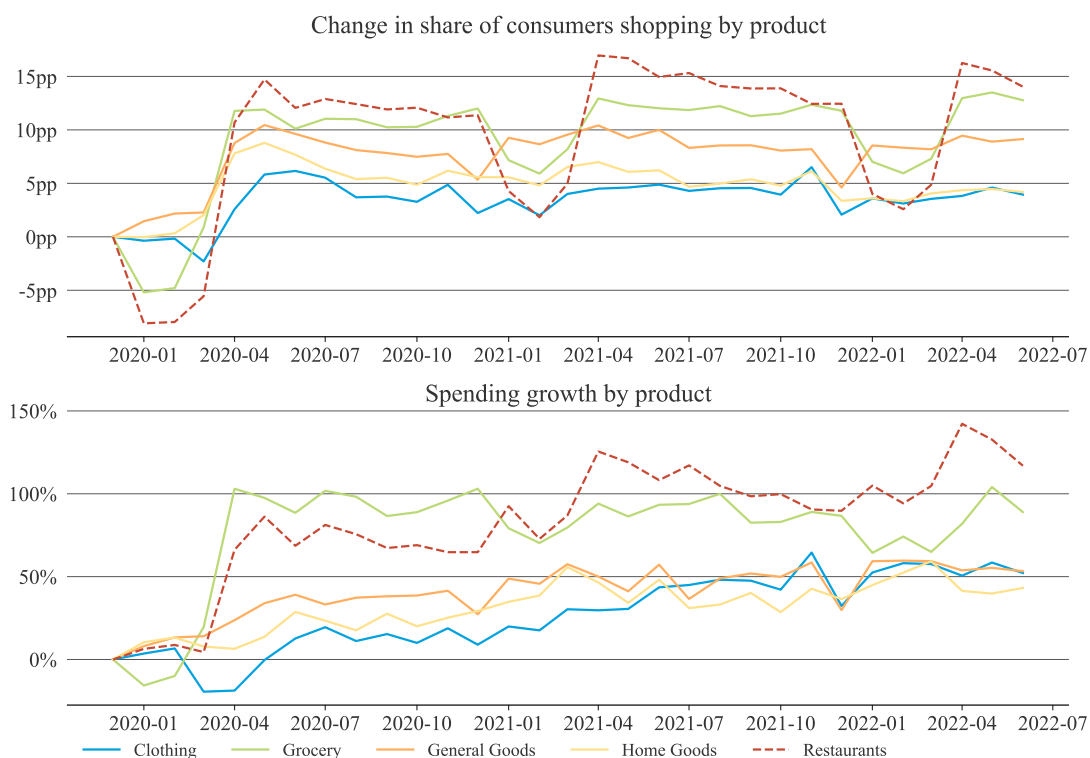
Notes: The top panel shows the percentage point change in the share of customers shopping online for products relative the share in the same month in 2019. For context, in February 2020 the share of customers shopping online for each product was 21 percent for clothing, 19 percent for grocery, 68 percent for general goods, 23 percent for home goods, and 49 percent for restaurants. The bottom panel shows the growth in spending relative to the same month in 2019. Winter declines in grocery and restaurant are because these products were already used more in winter months prior to the pandemic.

Sources: JPMorgan Chase.

For changes in online spending, we find that the pandemic led to the largest and most persistent increase in the adoption of newer online products, groceries and restaurants, as depicted in Figure A.10. The percentage of consumers using online channels for purchases of these products each month rose from 19 to 30 percent and 50 to 61 percent, respectively, between February and May 2020. Usage continues to widen to almost 15 percentage points for both products in the most recent data. In the case of groceries, this represents almost twice as much market penetration due to the pandemic. In contrast, fewer individuals increased their monthly use of more established online products – clothing, general goods, and home goods. This smaller increase is also declining over time. These results suggest that the fixed costs of adoption may play a role in the differential impact on new versus established online products, particularly for those products that are still in their introductory phase.

While grocery and restaurants also experience the strongest overall spending growth, our analysis

Figure A.10: Online spending growth



Notes: The top panel shows the percentage point change in the share of customers shopping online for products relative to the share in the same month in 2019. For context, in February 2020 the share of customers shopping online for each product was 21 percent for clothing, 19 percent for grocery, 68 percent for general goods, 23 percent for home goods, and 49 percent for restaurants. The bottom panel shows the growth in spending relative to the same month in 2019. Winter declines in grocery and restaurant are because these products were already used more in winter months prior to the pandemic.

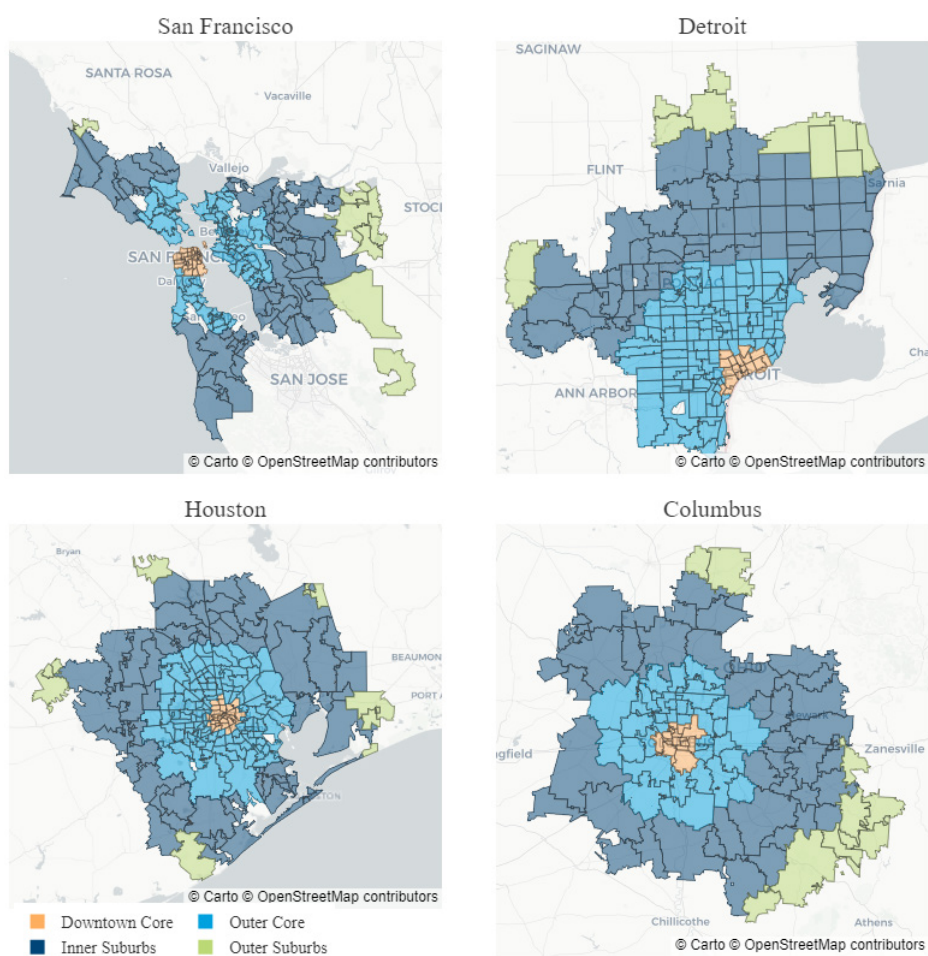
Sources: JPMorgan Chase.

shows that the overall increase in spending on established online products is also sustained and reflects a more frequent use of this channel by consumers who previously only made occasional online purchases (Relihan et al. 2020). As of June 2022, spending on general goods, home goods, and clothing was 53, 42, and 52 percent higher, respectively, compared to February 2020. These patterns may be influenced by factors such as inflation and an increase in overall spending due to saved income, but the recent online spending growth still significantly exceeds offline spending for these products. So much so, for instance, that more clothing is now purchased online than offline in our data as a result of the pandemic.

Given this increase, online retail is another technological shift driven by the pandemic that can affect the growth of establishments across space that we document. As with the earlier growth of online retail in the more tradable products, additional increases in spending online for general goods, home goods, and clothing could contribute to the continued depression in establishment growth in these products. But online groceries and restaurants, the products with the most growth during the pandemic, still have a large non-tradable component – in most cases fulfilling those online orders relies on local grocery stores and local restaurants. Thus, this type of online retail growth may actually be supportive of the local establishments growth we observe in suburban locations.

5.6 Additional supporting figures and tables

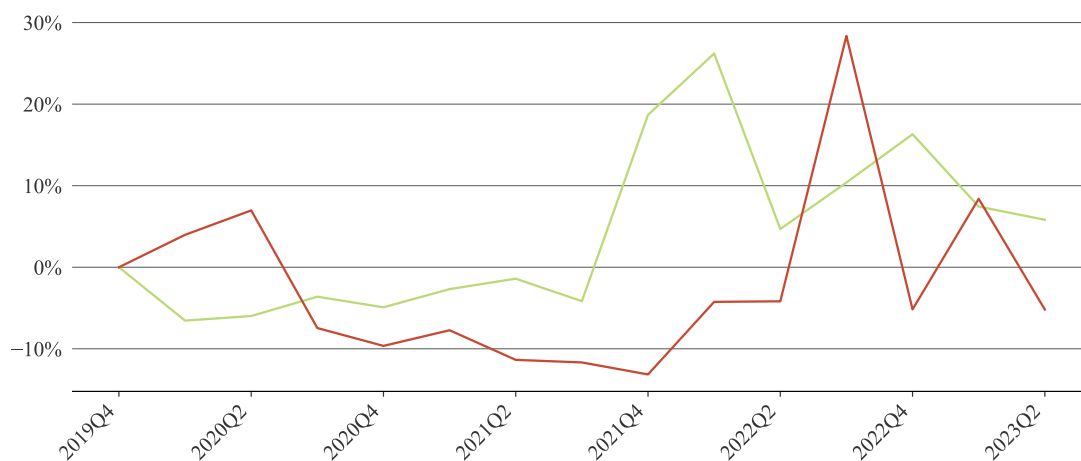
Figure A.11: Example city maps with region classification



Notes: This figure shows ZIP codes in each region of example cities defined with our scaled distance from 0 to 1, where ZIPs at the center have a distance of 0 and the furthest ZIPs have a distance of 1. As shown in these examples, the first three inner regions consistently create well-populated rings of ZIP codes. The outer suburbs are then comprised of a more sparse set of ZIP codes which are the farther from the city center but often disconnected from each other due to the shape of the CBSA boundary.

Sources: JPMorgan Chase.

Figure A.12: Growth in concentration of entries and exits



Notes: This figures shows the average growth in the HHI for entry (exit) of establishments across cities based on the share of entry (exit) for each zip code out of all entry (exit) in its respective city. Growth is relative to the same quarter in 2019.

Sources: JPMorgan Chase.

Table A.1: Alternative decomposition analysis of Q4 2019 - Q4 2021 establishment growth (%)

	1	2	3	4	5
Scaled Distance	0.107 (0.024)	0.052 (0.023)	0.091 (0.025)	0.039 (0.022)	0.02 (0.026)
Population Growth (%)		0.306 (0.065)		0.294 (0.065)	0.301 (0.11)
WFH residential exposure (%)			-0.017 (0.082)	-0.023 (0.079)	-0.162 (0.123)
WFH employment exposure (%)			-0.139 (0.059)	-0.123 (0.054)	-0.113 (0.054)
Online spending growth (%)					0.047 (0.024)
Offline spending growth (%)					-0.103 (0.107)
Non-white residents (%)					-0.025 (0.014)
Median income (\$1000s)					0.03 (0.024)
Mean Est. growth	2.82	2.82	2.82	2.82	2.82
Observations	3653	3653	3653	3653	3653
R-sq.	0.048	0.062	0.054	0.067	0.072
Adj. R-sq.	0.044	0.058	0.049	0.062	0.066

Notes: All models include CBSA fixed effects. Standard errors in parentheses and clustered by city.

Sources: JPMorgan Chase, U.S. Census Bureau American Community Survey and Longitudinal Employer-Household Dynamics Local Origin-Destination Employment Statistics.

Table A.2: Decomposition analysis of establishment growth over time

	Q2 2020	Q4 2020	Q2 2021	Q4 2021
Downtown Core	-0.364 (1.111)	-0.579 (0.715)	0.985 (1.819)	1.308 (0.84)
Population Growth (%)	0.304 (0.162)	0.284 (0.077)	0.384 (0.123)	0.318 (0.109)
WFH residential exposure (%)	-0.212 (0.114)	-0.317 (0.103)	-0.314 (0.103)	-0.226 (0.108)
WFH employment exposure (%)	-0.1 (0.044)	-0.091 (0.036)	-0.132 (0.052)	-0.106 (0.053)
Online spending growth (%)	0.032 (0.026)	0.013 (0.026)	-0.003 (0.035)	0.046 (0.024)
Offline spending growth (%)	0.091 (0.077)	0.014 (0.067)	0.013 (0.097)	-0.102 (0.107)
Non-white residents (%)	-0.045 (0.015)	-0.023 (0.014)	-0.007 (0.029)	-0.035 (0.018)
Median income (\$1000s)	0.04 (0.01)	0.033 (0.017)	0.057 (0.022)	0.037 (0.023)
Mean Est. growth	-3.44	-2.12	2.02	2.82
Observations	3651	3656	3649	3653
R-sq.	0.114	0.153	0.101	0.072
Adj. R-sq.	0.108	0.147	0.095	0.066

Notes: Growth is relative to the same quarter two years previous. All models include CBSA fixed effects. Standard errors in parentheses and clustered by city.

Sources: JPMorgan Chase, U.S. Census Bureau American Community Survey and Longitudinal Employer-Household Dynamics Local Origin-Destination Employment Statistics.

References

- Barnatchez, Keith, Leland D. Crane, and Ryan A. Decker.** 2017. “An Assessment of the National Establishment Time Series.” Board of Governors of the Federal Reserve System Finance and Economics Discussion Series 2017-10.
- Blevins, Jason R., Ahmed Khwaja, and Nathan Yang.** 2018. “Firm Expansion, Size Spillovers, and Market Dominance in Retail Chain Dynamics.” Management Science, 64(9): 4070–4093.
- Frey, William H.** 2022. “Table B. Major metro areas: Annual population growth, 2010-11 to 2019-21.” [Link](#). Accessed on 2022-12-15.
- Kleinman, Benny.** 2023. “Wage Inequality and the Spatial Expansion of Firms.” [Link](#). Accessed on 2024-03-18.
- Relihan, Lindsay, Marvin Ward, Chris Wheat, and Diana Farrell.** 2020. “The Early Impact of COVID-19 on Local Commerce: Changes in Spend Across Neighborhoods and Online.” Covid Economics, 1–28.